

MFA Developer Guide

Why use the Multi-Factor Authentication (MFA) Service?

The Shared Services & Components (SSC) Multi-Factor Authentication (MFA) Service allows applications to add 2nd-factor authentication for business scenarios that already support 1st-factor authentication.

Relevant scenarios for new usages of MFA:

- WK identity management / authentication provider that doesn't already support MFA
- Application using its own proprietary login authentication that doesn't already support MFA
- Application using a WK-internal authentication provider that doesn't already support MFA
- Application using a 3rd-party authentication provider that doesn't already support MFA
- Application using a 3rd-party authentication provider that does support MFA, but it is too expensive
- Application requiring lightweight re-validation of identity during a critical user-initiated workflow

Supported modes of MFA authentication:

- SMS (API requires only a phone number)
- Voice (API requires only a phone number)
- Email (API requires only an email address)
- WK Authenticator mobile app
- Google Authenticator, Microsoft Authenticator, and other TOTP authenticator mobile apps
- Temporary access code (generated via support portal)
- “Remember My Device” cookies
- WhatsApp (coming in 2026)

Other important MFA features:

- Lightweight MFA API can be called either via a REST API or through the MFA Client SDK (.NET and Java)
- Standard UI to make default integration simple
- Alternatively, optional custom UI can be implemented for application-specific requirements
- Multiple providers for SMS, voice, and email (to provide high availability/deliverability)
- Self-service admin portal

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Overview

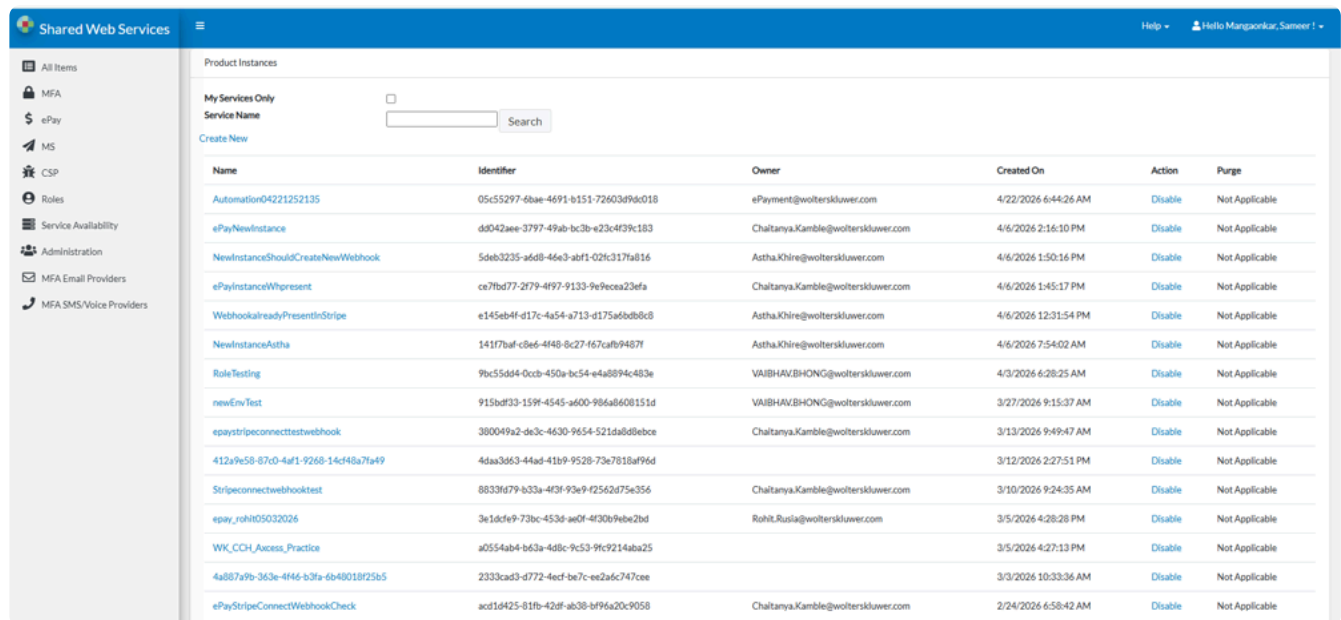
Using TechBV Shared Web Services is easy. Both administration and actual service can be accessed using REST API. Each service can be accessed using service instance name and access key. Both Basic and SharedKey authentication schemes are supported. You can see description of each method, its parameters and return values at `{Service_URL}/swagger`.

For example: refer `https://{base_url}/swagger`

Please refer to the [Environment Selection Guide on Confluence](#) for environment selection.

Configuration

Login to TechBV Shared Web Services portal using SSO. Click on MFA and "Create New":



The screenshot shows the 'Shared Web Services' portal interface. On the left is a sidebar with navigation links: All Items, MFA, ePay, MS, CSP, Roles, Service Availability, Administration, MFA Email Providers, and MFA SMS/Voice Providers. The main content area is titled 'Product Instances' and includes a search bar and a 'Create New' button. Below this is a table listing various service instances.

Name	Identifier	Owner	Created On	Action	Purge
Automation04221252135	05c55297-6bae-4691-b151-7260349d018	ePayment@wolterskluwer.com	4/22/2026 6:44:26 AM	Disable	Not Applicable
ePayNewInstance	dd042ae-3797-49ab-bc3b-e23c4f39c183	Chaitanya.Kamble@wolterskluwer.com	4/6/2026 2:16:10 PM	Disable	Not Applicable
NewInstanceShouldCreateNewWebhook	5deb3235-a6d8-46e3-abf1-02fc317fa816	Astha.Khire@wolterskluwer.com	4/6/2026 1:50:16 PM	Disable	Not Applicable
ePayInstanceWhPresent	ce7bd77-2f79-4f97-9133-9e9ecaa23efa	Chaitanya.Kamble@wolterskluwer.com	4/6/2026 1:45:17 PM	Disable	Not Applicable
WebhookAlreadyPresentInStripe	e145eb4f-d17c-4a54-a713-d175a6bdb8c8	Astha.Khire@wolterskluwer.com	4/6/2026 12:31:54 PM	Disable	Not Applicable
NewInstanceAstha	141f7ba8-c8e6-4f48-8c27-167caf9487f	Astha.Khire@wolterskluwer.com	4/6/2026 7:54:02 AM	Disable	Not Applicable
RoleTesting	9bc55d94-0ccb-450a-bc54-e4a894c483e	VAIBHAVBHONG@wolterskluwer.com	4/3/2026 6:28:25 AM	Disable	Not Applicable
newEnvTest	915bd33-159f-4545-a600-986a608151d	VAIBHAVBHONG@wolterskluwer.com	3/27/2026 9:15:37 AM	Disable	Not Applicable
epaystripeconnecttestwebhook	380049a2-de3c-4630-9654-521da8d8ebce	Chaitanya.Kamble@wolterskluwer.com	3/13/2026 9:49:47 AM	Disable	Not Applicable
412a9e58-87c0-4af1-9268-14c48a7fa49	4daa3d63-44ad-41b9-9528-73e7818af9fd		3/12/2026 2:27:51 PM	Disable	Not Applicable
Stripeconnectwebhooktest	8833fd79-b33a-4f3f-93e9-f2562d75e356	Chaitanya.Kamble@wolterskluwer.com	3/10/2026 9:24:35 AM	Disable	Not Applicable
epay_ruhit05032026	3e1dcfe9-73bc-453a-ae0f-4f30b9ebe2bd	Rohit.Rusla@wolterskluwer.com	3/5/2026 4:28:28 PM	Disable	Not Applicable
WK_CCH_Access_Practice	a0554ab4-b63a-4d8c-9c53-9fc9214aba25		3/5/2026 4:27:13 PM	Disable	Not Applicable
4a867a9b-363e-4f46-b3fa-6b48018f25a5	233cad3-d772-4ecd-be7c-ea2dc747cee		3/3/2026 10:33:36 AM	Disable	Not Applicable
ePayStripeConnectWebhookCheck	acd14425-81ff-42df-ab38-bf96a20c9058	Chaitanya.Kamble@wolterskluwer.com	2/24/2026 6:58:42 AM	Disable	Not Applicable

Enter unique MFA service name and specify to which domains MFA Emails can be sent. Once the MFA service instance is created, you can obtain access keys and configure text for SMS and Email messages. Default SMS and Email text for en-US culture is created when you create MFA service instance.

The screenshot shows the 'TAA Services' console with a sidebar on the left containing navigation links: All Items, MFA, xPay, SMS, CSP, Roles, Service Availability, Administration, MFA Email Providers, and MFA SMS/Voice Providers. The main area is titled 'Service Instance' and has three tabs: 'General' (selected), 'Provider(s) Configuration', and 'Registered Device(s) Configuration'. The 'General' tab contains the following fields:

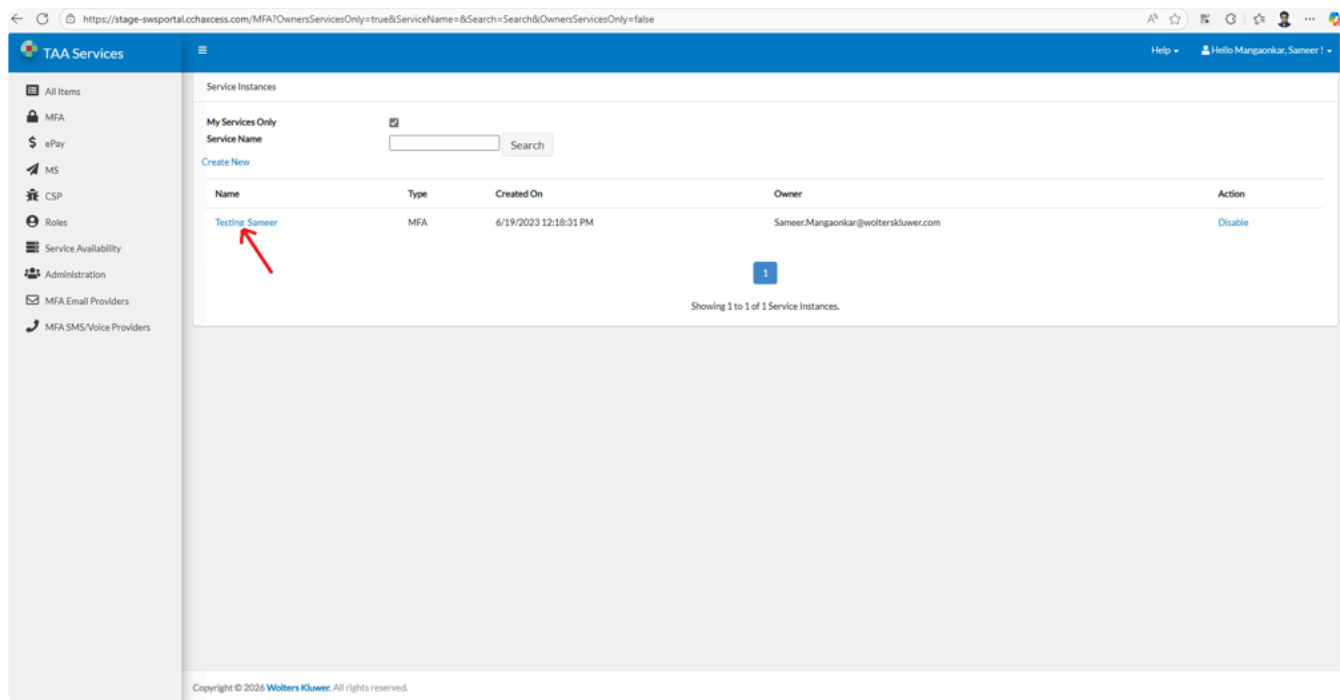
- Name:** A text input field.
- Sandbox:** A checkbox.
- Email Soft Disable:** A checkbox.
- Session Timeout (seconds):** A text input field.
- Allow Emails To:** A text input field.
- Compliance Mode:** A dropdown menu with 'None' selected.
- Tag Join:** A text input field with a placeholder 'Please enter join text for Tag value'.
- Default MFA Mode in Prompt UI:** A dropdown menu with 'None' selected.
- Whitelisted Client Users:** A text input field.
- Sandbox User Ids:** A text input field.
- Authorized OneClick Clients:** A text input field.
- Mandate App Authentication:** A checkbox.
- Authenticator Pairing Type:** A dropdown menu with 'Standard' selected.
- Trusted IP addresses:** A text input field.

At the bottom left of the form are 'Create' and 'Cancel' buttons. The footer of the console shows 'Copyright © 2020 Wolters Kluwer. All rights reserved.'

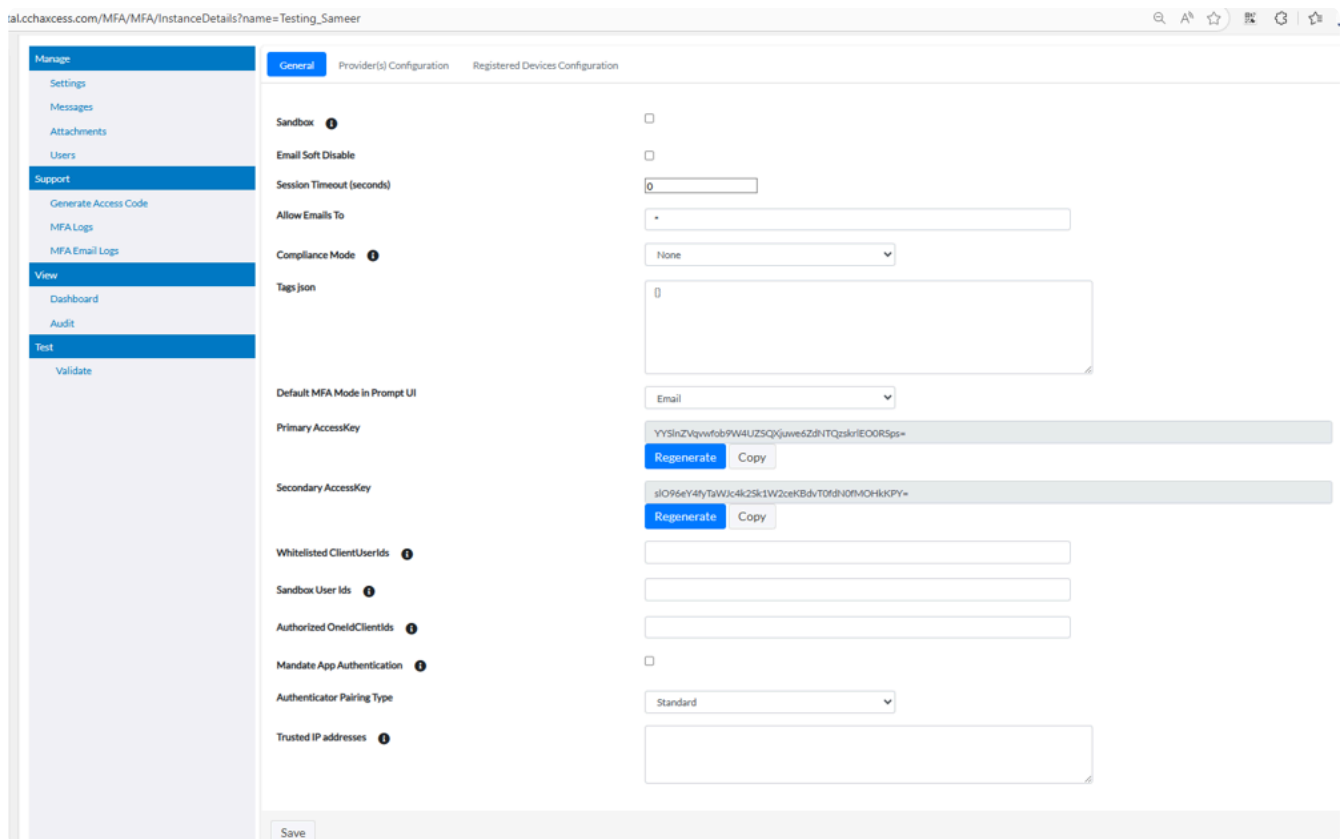
Field	Description
Name	Name of the service instance. A service instance represents an application registration of your application with the SSC service and acts as an access and configuration context. Must be at least 8 characters (alphanumeric and _ allowed, no spaces).
Division	Division of the WK product(s) using this service instance.
Product	List one or more WK products using this service instance. Used for itemized tracking on usage reports.
Billing Code	Used to calculate monthly chargeback of SMS costs. This will generally be a WK cost center. Please contact DL-DXG-SSC-Support@wolterskluwer.com with any questions.
Environment	Choose "Production" only for production service instances in a production environment.
Sandbox	This will be used for load testing. If set to true, the MFA response will include the verification code. Default value is false
Email Soft Disable	Allows email to be passed but does not display email as a user option unless all SMS/Voice providers are down and email providers are up
Session Timeout (seconds)	Time in seconds after which session expires
Allow Emails To	Restrict emails to specific domains

Compliance Mode	In Nist800-63b mode:- • Emails are not allowed for MFA. • SMS and Calls to Voip numbers are not allowed for MFA. • If the Voice Prompt Text in message template doesn't have <code>{{verificationCode}}</code> token then default Prompt text with <code>{{verificationCode}}</code> token will be used for voice MFA. In Audit Only mode:- • An Audit log entry will be generated when Call/SMS to Voip number or Email is selected for MFA. • If the Voice Prompt Text in message template doesn't have <code>{{verificationCode}}</code> token then default Prompt text with <code>{{verificationCode}}</code> token will be used for voice MFA.
Tags json	Used to store additional metadata about the service instance
Default MFA Mode in Prompt UI	Default selection in MFA Prompt UI
Whitelisted ClientUserIds	Whitelisted clientId list (separated by ';') to bypass MFA to support synthetic transaction monitoring agents.
Sandbox User Ids	Sandbox user id list and/or regex pattern list (separated by ';') to bypass sendSms, sendEmail & createMfaPrompt and return verification code.
Authorized OneIdClientIds	Authorized OneId clientId list (separated by ';'). Spaces will be removed. OneId tokens with these client Ids will be authorized for non-admin API calls for this service instance.
Mandate App Authentication	This will mandate the users to approve the Authenticator App Session requests with Passcode or Biometric Authentication if they configured in the Mobile.
Authenticator Pairing Type	Standard or IDaaS
Trusted IP addresses	For List of addresses, separate the IP address with comma. For a range of address, use <code>-</code> . For example: <code>192.168.1.1,192.168.1.2,192.168.1.1-200</code> . If the Client IP Address matches this IP address list, MFA will be bypassed.

After pressing "Create" button, you can see the newly created service instance and edit its configuration:



Obtain access keys and edit Allowed Domains. You can also regenerate access keys to invalidate previous keys:



Field	Description
Default MFA Mode in Prompt UI	Default choice on MFA prompt. None, Email, SMS, Voice, TOTP
Primary Access Key	Primary key for authorization

Secondary Access Key	Secondary key for authorization
----------------------	---------------------------------

Registered Devices Configuration:

For added security, API has a provision to include additional parameters such as IP and User Agent. Client can customize these configurations from Registered Devices Configuration.

Service Instance : Mfa_Stage

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General

Provider(s) Configuration

Registered Devices Configuration

Max Expiration (Days) ⓘ

30

Require IP ⓘ

☐

Require Same IP ⓘ

☐

Require User Agent ⓘ

☐

Require Same User Agent ⓘ

☐

Save

Field	Description
Max Expiration (Days)	The maximum number of days to allow for a device registration, to support policy-based security settings. Defaults to 30 if not set, based on NIST 800-63b AAL 1 requirement of re-authenticating the user every 30 days.
Require IP	If set to true, calls to verify a registered device must have the caller's IP sent or will be rejected. Additionally, calls to create an MFA session must include the clientIpAddress. Default value is false.
Require Same IP	If set to true, calls to verify a registered device must have the caller's IP sent and it must be the same as when registered device was created or will be rejected. Default value is false.
Require User Agent	If set to true, calls to verify a registered device must have the caller's user agent sent or will be rejected. Additionally, calls to create an MFA session must include the clientUserAgent. Default value is false.
Require Same User Agent	If set to true, calls to verify a registered device must have the caller's user agent sent and it must be the same as when registered device was created or will be rejected. Default value is false.

Add alternative text for SMS and Email. Message name and culture will be used as parameters when calling MFA service methods:

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Message Details

Create New

Name: Default

Add New Culture

Import

Export

Name	Action	Action
en-US	Edit	Delete
it	Edit	Delete

Place `{{verificationCode}}` tag where verification code should be added in actual SMS or Email text:

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Message Details

Name

Default

Culture Code

en-US

SMS Text

Your sign-in verification code is: {{verificationCode}}.
For your security, please don't forward this message.

Twilio SMS From Phone No.

Plivo SMS From Phone No.

Voice Prompt Text

Thank you for using the Wolters Kluwer authentication service. Please press the # sign to complete your authentication.

Voice Completion Text

Your authentication has been successfully completed. Goodbye!

Voice Failure Text

I am sorry we were not able to complete your authentication. Goodbye!

Twilio Voice From Phone No.

Plivo Voice From Phone No.

From

noreply@wolterskluwer.com

Email Subject

Verify your sign-in

Email Text Body

Please use the following sign-in verification code when prompted: {{verificationCode}}.
For your security, please don't forward this message.

Email Default Attachments

Email Html Body

<html><body>Please use the following sign-in verification code when prompted: {{verificationCode}}.
For your security, please don't forward this message.</body></html>

Time zone

Access Code Helper Text

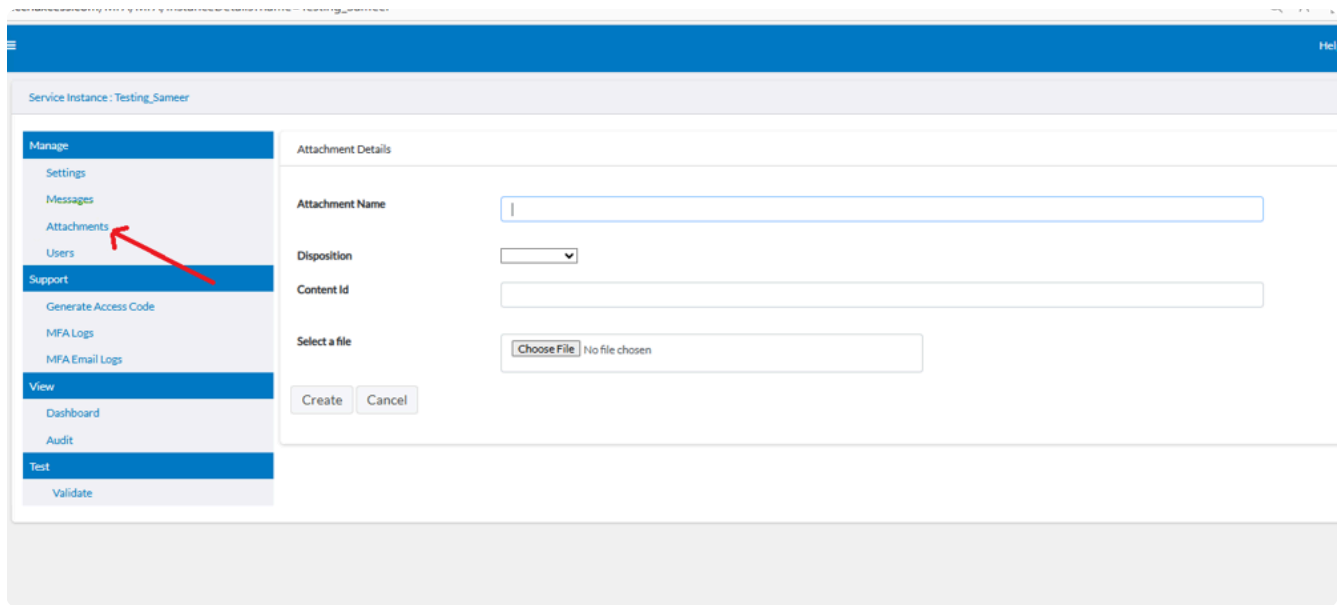
"Sms Text", "Email Subject", "Email Text Body" and "Email Html Body" serve as Liquid template.

Specify comma separated attachment names in "Email Default Attachments": these will be attached with each Email (for this message culture) without specifying the names in each SendEmail method call.

`messageSubstitutions` parameter in `sendEmail` and `sendSms` MFA methods is json containing data for Liquid template.

To create attachments for your HTML emails, click on "attachments", then on "Create New".

Enter attachment name, content Id, disposition then select file to upload and click create.



Service Instance: Testing_Sameer

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Attachment Details

Attachment Name

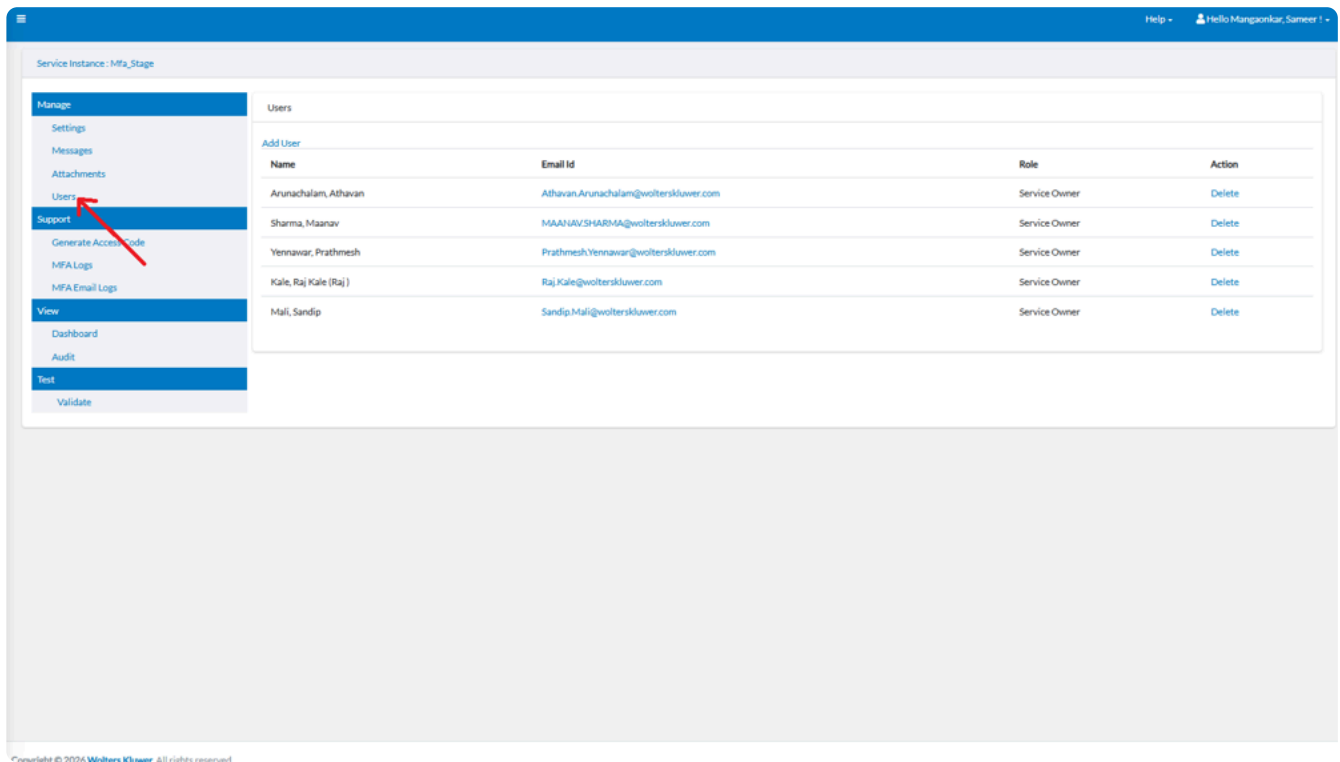
Disposition

Content Id

Select a file No file chosen

Content ID must match to cid attribute of img html tag in order for images to be displayed inline. Attachment name will be used as one of elements in "attachments" parameter in sendEmail MFA method.

To add/delete users for current Service Instance, click on Users:



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Users

[Add User](#)

Name	Email Id	Role	Action
Arunachalam, Athavan	Athavan.Arunachalam@volterskluwer.com	Service Owner	Delete
Sharma, Maanav	MAANAV.SHARMA@volterskluwer.com	Service Owner	Delete
Yennavar, Prathmesh	Prathmesh.Yennavar@volterskluwer.com	Service Owner	Delete
Kale, Raj (Kale (Raj))	Raj.Kale@volterskluwer.com	Service Owner	Delete
Mail, Sandip	Sandip.Mail@volterskluwer.com	Service Owner	Delete

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Adding Users

Click on the Add User link, it will take you to the Find user page. Now tick the checkbox of corresponding user and click on Select User(s) button. Next step is to select Role for the user.

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Users

Step 1
Select User(s)

Step 2
Select Role

Name/Email Starts with

Select	Email Id	Given Name	Surname	Display Name
☐	Chaitanya.Busam@wolterskluwer.com	Chaitanya	Busam	Busam, Chaitanya
☐	Chaitanya.Gosavi@wolterskluwer.com	Chaitanya	Gosavi	Gosavi, Chaitanya
☒	Chaitanya.Kamble@wolterskluwer.com	Chaitanya	Kamble	Kamble, Chaitanya Kamble (Chaitanya)
☐	chaitanya.said@wolterskluwer.com	Chaitanya	Said	Said, Chaitanya
☐	Chaitanya.Vegesna@wolterskluwer.com	Chaitanya	vegesna	vegesna, Chaitanya

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Showing 1 to 5 of 5 Users

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Step 1
Select User(s)

Step 2
Select Role

Select Role

Select Role
Product Support Representative
Product Support Supervisor
Service Owner

For users, there are three roles defined as follows:

- **Service Owner** - Has full access to the service instance
- **Product Support Representative** - Has access to only Support section which includes MFA Logs/MFA Email logs
- **Product Support Supervisor** - Has access to Users and SMS Logs

For a detailed Support Role Guide refer to [User & Developer Guides — Tech BV Shared Services & Components \(SSC\)](#).

Generate Access Code: If there is an outage from user end, and they are not able to login using any of the MFA methods, an admin can generate an access code for 24 hours using which the user can login.

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Access Code Generator

User Id

Sameer

Notes

Outage at customer

Generate

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The following access code has been generated for the user id : Sameer

430351

The code expires in 24 Hours

MFA Logs:

This section displays MFA Logs between timestamp for a particular user.

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MFA Logs

From (UTC)

01/28/2026 7:49 AM

To (UTC)

02/05/2026 7:49 AM

User Id

MFAChaitanyaainst

Search

Download Logs

MFA Initiated Timestamp	Authentication Type	Status	Client IPAddress	Delivery Info
2/3/2026 7:23:55 AM	Email	Succeeded		Test email was sent to chaitanya.kamble@wolterskluwer.com using SendGrid email provider
2/3/2026 7:23:55 AM	Email	Succeeded		Test email was sent to chaitanya.kamble@wolterskluwer.com using Mailgun email provider
2/3/2026 6:38:11 AM	MfaPrompt	InProgress	123.123.234.234	MfaPrompt Session timeout: 00:08:39.7728585
2/3/2026 6:38:10 AM	MfaPrompt	InProgress	123.123.234.234	MfaPrompt Session timeout: 00:08:39.4184943
2/3/2026 6:24:23 AM	MfaPrompt	InProgress	123.123.234.234	
2/3/2026 6:15:21 AM	Voice	Succeeded	123.123.234.234	[{"ServiceType": "MFA", "Accountid": "MAN2M3ZWRMME1ZJ13ZJ", "FromNumber": "15103193231", "ToNumber": "918378989154", "Requestid": "B8855f46-e429-4516-aa26-576c4e5bf9b9", "Callid": "B8855f46-e429-4516-aa26-576c4e5bf9b9"}]

MFA Email Logs:

This section displays Email logs sent between timestamp for a particular user. You can also send test email using this section.

Service Instance : MFAsanitytest

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MFA Email Logs

From (UTC)

01/28/2026 7:54 AM

To (UTC)

02/05/2026 7:54 AM

User Id

Email Id

username@wolterskluwer.com

Event

All

Search

Send Test Email

Timestamp (UTC)	Event	Email	User Id	Reason
2/4/2026 7:53:32 AM	delivered	chaitanya.kamble@wolterskluwer.com	username@wolterskluwer.com	
2/4/2026 7:53:31 AM	processed	chaitanya.kamble@wolterskluwer.com	username@wolterskluwer.com	

1

Dashboard:

This section displays a counter for the requests sent over the 24 hours and 30 days' period.

Service Instance : Mfa_Stage		
Manage	Count specified below is the number of MFA event logs generated by last 1000 sessions only. Each session can have more than 1 event logs. For detailed log analysis, please refer MFA Logs from support section.	
Settings	Total Requests in Last 24 Hrs :	2
Messages	Total Failed Requests in Last 24 Hrs :	0
Attachments	Total Requests in Last 30 Days :	5
Users	Total Failed Requests in Last 30 Days :	0
Support		
Generate Access Code		
MFA Logs		
MFA Email Logs		
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Validate		

Audit:

This section allows user to download MFA session events for last 100 requests and for last 30 days.

Service Instance : Mfa_Stage		
Manage	Audit Logs	
Settings	Audit Logs are retained for online for 30 Days The exported csv for last 30 days will have MFA event logs generated by last 1000 sessions only. For detailed log analysis, please refer MFA Logs from support section. Download CSV for last 100 requests Download CSV for last 30 Days	
Messages		
Attachments		
Users		
Support		
Generate Access Code		
MFA Logs		
MFA Email Logs		
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You can see the list of registered devices for your service owner account on Profile page. Here other profile details such as Email, Display name, Primary and Secondary access keys, JWT auth token and paired authenticator software's are displayed.

POST

/api/mfa/sendEmail

Initiates an email authentication request

Sends the user an email based on the configured message with a one-time password. The application should prompt the user to enter their code, and then verify that code with the returned MFA Session ID. Requires Service Instance authentication.

Parameters

Try it out

No parameters

Request body

application/json-patch+json

The email parameters to send

Example Value | Schema

```

{
  "messageName": "Default",
  "cultureCode": "en-US",
  "emailAddress": "username@liteskuser.com",
  "clientIpAddress": "123.123.234.234",
  "clientId": "username@liteskuser.com",
  "clientUserAgent": "Mozilla/5.0 (Windows NT 10.0; MSM4; Trident/7.0; rv:11.0) like Gecko",
  "attachments": [
    ""
  ],
  "messageSubstitutions": "( \\user\\: ( \\state\\: \\GA\\, \\address\\: \\123 Street\\) )"
}

```

Responses

Code	Description	Links
200	OK	No links
	Media type text/plain Content Accept header Example Value Schema <pre> { "mfaId": "8DCA3EA772568C03F638825F4888DA58", "status": "InProgress", "lockoutStatus": "None", "verificationCode": "" } </pre>	
400	Bad Request	No links
	Media type text/plain Example Value Schema Product User	
500	Internal Server Error	No links
	Media type text/plain Example Value Schema Product User	

Using the ID of MFA session and verification code that you should receive an Email, you can verify the code:

POST /api/mfa/sendSms Initiates an SMS authentication request

POST /api/mfa/sendVoice Initiates a voice authentication request

POST /api/mfa/verifyCode Verifies the code for the given MFA session

Implementation Notes
Verifies a PIN code that the user entered after an SMS or Email authentication Requires Service Instance authentication.

Response Class (Status 200)

Response Content Type: application/json

Parameters

Parameter	Value	Description	Parameter Type	Data Type
request	<pre>{ "mfaId": "b3f545afb9284711aa276caa56780a4a", "code": "696413" }</pre>	The code and MFA session to verify	body	Model Model Schema

Parameter content type: application/json

Try it out! [Hide Response](#)

Curl

```
curl -X POST --header 'Content-Type: application/json' --header 'Accept: application/json' --header 'Authorization: Bearer AlexMfaSe
'mfaId': 'b3f545afb9284711aa276caa56780a4a',
'code': '696413'
}' 'https://eastus-taasdemo5-webapi.azurewebsites.net/api/mfa/verifyCode'
```

Request URL

```
https://eastus-taasdemo5-webapi.azurewebsites.net/api/mfa/verifyCode
```

Response Body

```
"success"
```

Response Code

```
200
```

Response Headers

```
{
  "pragma": "no-cache",
  "strict-transport-security": "max-age=31536000",
  "content-encoding": "gzip",
  "cache-control": "no-cache",
  "date": "Thu, 16 Jun 2016 16:14:56 GMT",
  "vary": "Accept-Encoding",
  "content-type": "application/json; charset=utf-8",
  "arr-disable-session-affinity": "true",
  "content-length": "128",
  "expires": "-1"
}
```

POST /api/mfa/getStatus Gets the status of the given MFA Session ID

PUT /api/mfa/getTotpKey Creates new user key.

POST /api/mfa/checkIfTotpKeyExists Checks if Totp key exists.

POST /api/mfa/setTotpEnabled Set Enabled flag.

POST /api/mfa/verifyTotp

All of the methods of administrator management API are available via service web site as REST API. For authentication you should use service owner Email Id + ':' (colon character) + access key as API key. You can obtain this information at https://{base_url}/ServiceOwner

- `ServerBaseUri` is the base URL of the MFA service website. Refer to the [Environment Selection Guide](#) if you need to target a different environment.
- `ServiceInstanceName` is the name of service instance you created in the portal or via management API.
- `ServiceInstanceKey` is the access key of service instance you created in the portal or via management API.

General security considerations

[!NOTE] When using `SharedKey` authentication (the only option when using client SDK), the difference between the caller (SDK code) and the server time should be within 15 minutes.

[!NOTE] MFA session will time-out in 5 minutes (by default, according to **Session Timeout** of the service instance). If the user doesn't complete MFA session within 5 minutes, the response will be "fail".

[!NOTE] If the same user (identified by provided **clientUserId**) uses MFA more than 5 times within 15 minutes, the result will be "fail". This is to defend against brute force attack on the 6 digit One-Time Passcode.

MFA Prompt Dialog

The MFA Prompt Dialog provides a standard user interface that applications can use to allow users to perform MFA.

To integrate this way, the application passes in the MFA parameters on how to contact the user (all optional, we'll show radio buttons accordingly) as well as a `redirectUrl` for continuation and gets back a **mfaUiUrl** with an embedded session ID. Then the application directs the user to **mfaUiUrl**. Once the user completes the steps of MFA, either because they had chosen "remember my device" earlier or through actually completing the challenge, it'll redirect back to the application's **redirectUri** with an **mfaCompletionId** on the querystring. The application makes a separate call to `validateMfaCompletion` and can ensure that the **mfaCompletionId** was valid, and gets back some information it can choose to record in audit logs on how the user completed the challenge.

In the current release, the following features of MFA are supported with `createMfaPrompt`:

1. Email
2. SMS
3. Voice Call, with SignalR detection of the out-of-band authentication being completed
4. Authenticator apps paired with a QR code generated by the API `generateTotpKey` where the `userId` for the TOTP key matches the `clientUserId` of the `createMfaPrompt` call
 - Authenticator app using a push notification and allowing user to approve on their smart phone, with SignalR detection of the out-of-band authentication
 - Google Authenticator using a code generated by the app

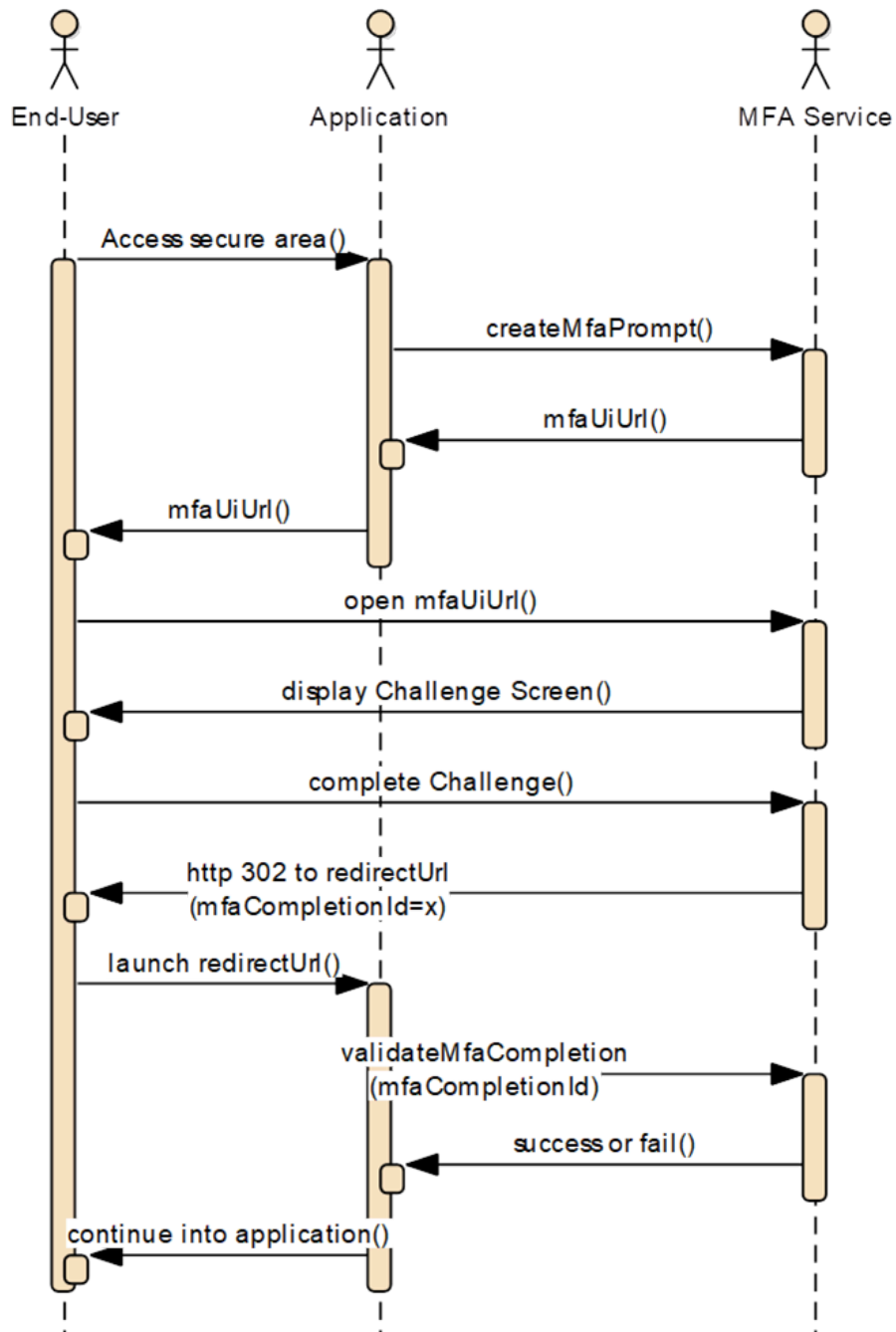
[!NOTE] MFA Prompt will display Authenticator App option (Wolters Kluwer Authenticator / Google Authenticator) only if it is verified before consuming MFA Prompt. Applications consuming Device Pairing Prompt / Regular Auth Pairing Prompt will be verifying the pairing as part of Prompt UI.

Applications who use `"/api/mfa/generateTotpKey"` to generate QR Code for the client user id, should be initiating `/api/mfa/startTotpSession` and Verifying the TOTP using `/api/mfa/verifyTotp` before using the MFA Prompt.

5. "Remember Me" where the user was given the option to remember their device on initial sign-on, and optionally sliding expiration where their device cookie is renewed each time they sign in, thus only requiring MFA if they have not signed on at all from a device for > 30 (or n) days
6. Access Code - If the user ID had an access code generated through a support using TechBV SWS Portal or by your application using the `createAccessCode` API then the MFA prompt will automatically display a link for the user to use it.

Here is a picture of the integration:

sd Dynamic View



Create MFA Prompt session

Call `client.CreateMfaPrompt()` method to create MFA Prompt session.

Request Body Parameters:

Parameter	Description
<code>messageName</code>	Name of the message template to be used for MFA modes that use message template.

<code>cultureCode</code>	Culture code to be used for MFA modes that use culture code. (Currently supported CC : ar- Arabic, de- German, es: Spanish, fr: French, id-ID: Indonesia, it: Italian, ja: Japanese, nl: Dutch, pt: Portuguese, ru: Russian, sv: Swedish, ta: tamil, th: thai, vi: Vietnamese, zh: chinese)
<code>emailAddress</code>	Email address to be used for email MFA mode. Multiple E-mail id(s) can be separated by " "
<code>smsCountryCode</code>	Sms country code(s) SMS MFA mode. Multiple numbers can be separated by " " (called as vertical bar or pipe line). This should match the SMS phone numbers count.
<code>smsPhoneNumber</code>	Sms phone number(s) SMS MFA mode. Multiple numbers can be separated by " "
<code>allowInternationalCalls</code>	Allow international calls flag for SMS and voice MFA modes.
<code>voiceCountryCode</code>	Voice country code Voice MFA mode. Multiple numbers can be separated by " "
<code>voicePhoneNumber</code>	Voice phone number Voice MFA mode. Multiple numbers can be separated by " "
<code>attachments</code>	An optional array of attachment names used in Email MFA mode.
<code>messageSubstitutions</code>	Message substitutions Json object used in Email and SMS MFA modes.
<code>allowRegisteredDevice</code>	True to allow using registered device MFA mode. Default is true.
<code>createDeviceExpirationTimespan</code>	Expiration to use in create registered device method.
<code>verifyDeviceExpirationTimespan</code>	Expiration to use in verify registered device method.
<code>clientIpAddress</code>	Client IP address to use to create MFA session for all MFA modes.
<code>clientUserAgent</code>	Client user agent to use to create MFA session for all MFA modes.
<code>clientUserId</code>	Client user id to use to create MFA session for all MFA modes.
<code>redirectUri</code>	URL to redirect to after completion of MFA.
<code>redirectOnErrorUri</code>	URL to redirect to in case of error in MFA Prompt.
<code>bannerAttachmentName</code>	This is the attachment name of the logo image file which is added as attachments in the MFA instance.
<code>stylesheetAttachmentName</code>	This is the attachment name of the style sheet file which is used in the MFA prompt UI.
<code>trustedIpAddress</code>	IP addresses for which MFA authentication will be bypassed.

Response Class (Status 200 OK):

Field	Description
mfaUiUrl	URL for MFA Prompt dialog. This URL is valid for 1 minute

Within 1 minute the client browser should be redirected to the URL specified by response (mfaUiUrl).

The user will have 5 minutes (**Session Timeout** value in Service instance settings) to complete MFA.

MFA Prompt UI Custom Stylesheet

The MFA Prompt UI provides a standard user interface that applications can use to allow users to perform MFA. The color scheme of the UI can be customized according to your application. The colors of the HTML elements can be changed by overriding the CSS classes.

Please refer the below code sample. The highlighted values can be changed along with other attributes that is part of the style classes. Most of the name of CSS classes are self-explanatory. You can also identify the elements by inspecting the UI using the browser's developer tools.

```

1  body, html {
2      font-size: 16px;
3      line-height: 1.4;
4  }
5
6  .wk-mfa-panel .wk-mfa-header {
7      background: #24744c;
8  }
9
10 .wk-mfa-panel .wk-mfa-frame .wk-mfa-frame-header {
11     background: #24744c;
12 }
13
14 .btn-primary {
15     color: #fff;
16     background-color: #24744c;
17     border-color: #357ebd;
18 }
19
20 .btn-primary:hover,
21 .btn-primary:focus,
22 .btn-primary:active,
23 .btn-primary.active,
24 .open .dropdown-toggle.btn-primary {
25     color: #fff;
26     background-color: #24744c;
27     border-color: #357ebd;
28 }
29
30 .btn-primary.disabled,
31 [disabled].btn-primary,
32 fieldset[disabled] .btn-primary,
33 .btn-primary.disabled:hover,
34 [disabled].btn-primary:hover,
35 fieldset[disabled] .btn-primary:hover,
36 .btn-primary.disabled:focus,
37 [disabled].btn-primary:focus,
38 fieldset[disabled] .btn-primary:focus,
39 .btn-primary.disabled:active,
40 [disabled].btn-primary:active,
41 fieldset[disabled] .btn-primary:active,
42 .btn-primary.disabled.active,
43 [disabled].btn-primary.active,
44 fieldset[disabled] .btn-primary.active {
45     background-color: #24744c;
46     border-color: #357ebd;
47 }
48
49 :not([disabled]).wk-button:hover,
50 :not([disabled])[wk-button]:hover,
51 button:not([disabled]):hover {
52     color: #fff;
53     background-color: #24744c;
54     border-color: #357ebd;

```

```

55 }
56
57 .wk-mfa-panel .wk-mfa-frame .wk-mfa-frame-body .wk-mfa-options .list-group.phone-number-list .selected-email-item {
58     color: #fff;
59     background-color: #24744c;
60 }
61
62 .wk-mfa-panel .wk-mfa-frame .wk-mfa-frame-body .wk-mfa-options .list-group.phone-number-list .selected-item {
63     color: #fff;
64     background-color: #24744c;
65 }

```

Once the stylesheet is customized for your application, please save it as ".css" file and ensure that the CSS is minified (i.e., remove unnecessary whitespace, comments, and formatting) which can be uploaded into TechBV SWS Portal. Please refer the [Configuration](#) section in this document, to upload as an attachment to your MFA service instance.

SERVICE INSTANCE : MFA_CUSTOM_UI_DEMO

Manage

Settings

Messages

Attachments

Users

Support

Generate Access Code

MFA Logs

MFA Email Logs

View

Dashboard

Audit

Test

Validate

ATTACHMENT DETAILS

Attachment Name

MyCustomStyle

Disposition

Attachment

Content Id

MyCustomStyle

Select a file

C:\Users\Raja.Ranganathan\Desktop\brasilStyle.css

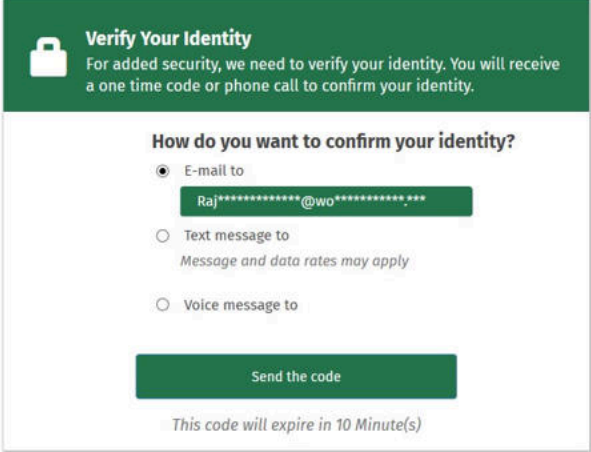
Browse...

Create

Cancel

When calling the method `client.CreateMfaPrompt()` as mentioned in the [Create MFA Prompt session](#) section, pass the `stylesheetAttachmentName` attribute value as the one you added as stylesheet attachment in your MFA Service Instance.

When rendering the MFA Prompt Dialog UI, the customized style will be in effect. Sample Image:



Verify Your Identity
For added security, we need to verify your identity. You will receive a one time code or phone call to confirm your identity.

How do you want to confirm your identity?

☒ E-mail to
Raj*****@wo*****

☐ Text message to
Message and data rates may apply

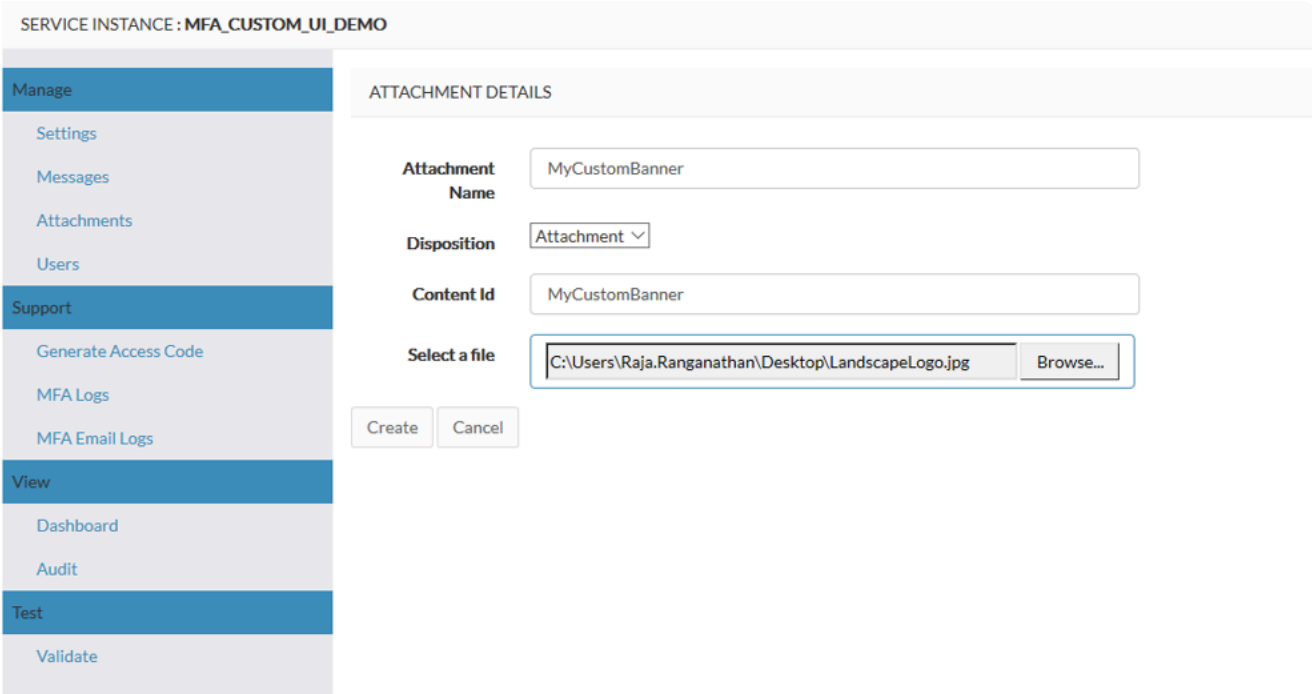
☐ Voice message to

Send the code

This code will expire in 10 Minute(s)

MFA Prompt UI Custom Banner Logo Image

The standard MFA Prompt UI doesn't display a banner image above the MFA dialog. If your application requires a banner image, a jpg/png image can be added to your MFA service instance as an attachment. The Image aspect ratio should be 16:9 (Landscape Mode) to display the image correctly. By default, the image will be stretched out in the place holder.



SERVICE INSTANCE : MFA_CUSTOM_UI_DEMO

Manage

- Settings
- Messages
- Attachments
- Users

Support

- Generate Access Code
- MFA Logs
- MFA Email Logs

View

- Dashboard
- Audit

Test

- Validate

ATTACHMENT DETAILS

Attachment Name MyCustomBanner

Disposition Attachment

Content Id MyCustomBanner

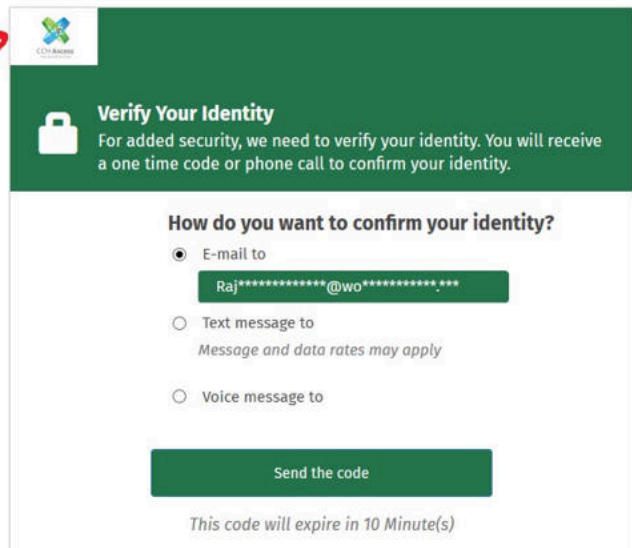
Select a file C:\Users\Raja.Ranganathan\Desktop\LandscapeLogo.jpg **Browse...**

Create **Cancel**

When calling the method `client.CreateMfaPrompt()` as mentioned in the [Create MFA Prompt session](#) section, pass the `bannerAttachmentName` attribute value as the one you added as banner attachment in your MFA Service Instance.

When rendering the MFA Prompt Dialog UI, the customized banner will be shown. Sample Image:

Custom Logo



Verify Your Identity
For added security, we need to verify your identity. You will receive a one time code or phone call to confirm your identity.

How do you want to confirm your identity?

☒ E-mail to
Raj*****@wo*****

☐ Text message to
Message and data rates may apply

☐ Voice message to

Send the code

This code will expire in 10 Minute(s)

MFA Prompt UI Redirect Scenarios

Successfully completes MFA

After the user performs MFA, the MFAPrompt dialog redirects the browser to **redirectUri** URL with added query string parameter:

- **mfaCompletionId** : Id of completion of the MFA Prompt Dialog.

User denies the MFA

In case of user denying the MFA attempt in Wolters Kluwer Authenticator App, the MFAPrompt dialog redirects the browser to **redirectOnErrorUri** URL with the querystring **errorCode=Denied** . If **redirectOnErrorUri** is empty, it will redirect to **redirectUri** with the query string **errorCode=Denied**

The MFA Session Time-out

If the user didn't complete the MFA within the configured MFA session timeout interval, the MFAPrompt dialog redirects the browser to **redirectOnErrorUri** URL with the querystring **errorCode=SessionExpired** . If **redirectOnErrorUri** is empty, it will redirect to **redirectUri** with the query string **errorCode=SessionExpired**

In MfaPromptUI, in some cases, user may be prompted to click a link to be redirected back.

The MFA user locked-out (Client User Id)

If the user makes 5 consecutive failed attempts of MFA in 15 mins, the MFAPrompt dialog redirects the browser to **redirectOnErrorUri** URL with the querystring **errorCode=UserLockedOut** . If **redirectOnErrorUri** is empty, it will redirect to **redirectUri** with the query string **errorCode=UserLockedOut**

In MfaPromptUI, in some cases, user may be prompted to click a link to be redirected back.

Too Many Attempts of MFA (Client User Id)

If the user makes 5 consecutive attempts of MFA in 1 min, the MFAPrompt dialog redirects the browser to **redirectOnErrorUri** URL with the querystring **errorCode=TooManyAttempts** . If **redirectOnErrorUri** is empty, it will redirect to **redirectUri** with the query string **errorCode=TooManyAttempts**

In MfaPromptUI, in some cases, user may be prompted to click a link to be redirected back.

Unknown Error

The MfaPrompt will redirect back to **redirectUri** with the query string `errorCode=MFAServiceUnknownError` if an unexpected error (such as database connection failure) received from MFA Service.

Bad Request

The MfaPrompt will redirect back to **redirectUri** with the query string `errorCode=MFABadRequest` if the provided input is incorrect (such as modified MfaId).

Verify MFA Prompt session

Call `client.ValidateMfaCompletionToken()` method.

Query Parameters:

Parameter	Description
<code>mfaCompletionId</code>	MFA Prompt Dialog completion Id.
<code>clientId</code> (optional)	Client user id used in CreateMfaPrompt() method.

Response Class (Status 200 OK):

Field	Description
<code>mfaId</code>	MFA Session Id of the MFA session that succeeded in MFA Prompt Dialog.
<code>mfaMode</code>	MFA Mode of the MFA session that succeeded in MFA Prompt Dialog.
<code>status</code>	MFA Status of the MFA session that succeeded in MFA Prompt Dialog.

SendEmail:Initiates an Email authentication request

Call `api/mfa/sendEmail` to initiates an Email authentication request.

Request Body Parameters:

Parameter	Description
<code>messageName</code>	Name of the message template to be used for MFA modes that use message template.
<code>cultureCode</code>	Culture code to be used for MFA modes that use culture code. (Currently supported CC : ar- Arabic, de- German, es: Spanish, fr: French, id-ID: Indonesia, it: Italian, ja: Japanese, nl: Dutch, pt: Portuguese, ru: Russian, sv: Swedish, ta: tamil, th: thai, vi: Vietnamese, zh: chinese)
<code>emailAddress</code>	Email address to be used for email MFA mode. Multiple E-mail id(s) can be separated by " "
<code>clientIpAddress</code>	Client IP address to use to create MFA session for all MFA modes.

clientUserAgent	Client user agent to use to create MFA session for all MFA modes.
clientId	Client user id to use to create MFA session for all MFA modes.
attachments	An optional array of attachment names used in Email MFA mode.
messageSubstitutions	Allows recipient-specific placeholder values to be dynamically replaced in a shared email template, enabling personalized bulk email delivery using a single Message Profile and Template.

Response Class (Status 200 OK):

Field	Description
mfaId	Unique identifier of the MFA session created for the email authentication request.
status	Current status of the email authentication request and verification workflow.
lockoutStatus	Indicates whether a user is currently blocked from completing MFA authentication due to security policies or exceeded verification attempts

SendSMS:Initiates an SMS authentication request

Call `api/mfa/sendSms` to initiates an SMS authentication request.

Request Body Parameters:

Parameter	Description
messageName	Name of the message template to be used for MFA modes that use message template.
cultureCode	Culture code to be used for MFA modes that use culture code. (Currently supported CC : ar- Arabic, de- German, es: Spanish, fr: French, id-ID: Indonesia, it: Italian, ja: Japanese, nl: Dutch, pt: Portuguese, ru: Russian, sv: Swedish, ta: tamil, th: thai, vi: Vietnamese, zh: chinese)
countryCode	Sms country code(s) SMS MFA mode. Multiple numbers can be separated by " " (called as vertical bar or pipe line). This should match the SMS phone numbers count.
PhoneNumber	Sms phone number(s) SMS MFA mode. Multiple numbers can be separated by " " (called as vertical bar or pipe line).
allowInternationalCalls	Allow international calls flag for SMS and voice MFA modes.(deprecated)
clientIPAddress	Client IP address to use to create MFA session for all MFA modes.
clientUserAgent	Client user agent to use to create MFA session for all MFA modes.
clientId	Client user id to use to create MFA session for all MFA modes.

messageSubstitutions	Allows recipient-specific placeholder values to be dynamically replaced in a shared email template, enabling personalized bulk email delivery using a single Message Profile and Template.
allowVoip	Determines whether MFA SMS or voice call verification can be sent to a VoIP phone number; when disabled, SMS and call delivery to VoIP numbers is not permitted.
product	List one or more WK products using this service instance. Used for itemized tracking on usage reports. Can enter upto 256 characters.
division	Division of the WK product(s) using this service instance. Allowed values: TAA, CPESG, LR, FCC, HEALTH, DXG, OTHER.
billingCode	Used to calculate monthly chargeback of SMS costs. This will generally be a WK cost center. Can enter upto 256 characters.

Note: These fields (product, division & billingCode) are collectively optional but conditionally required—if any one of product, division, or billingCode is provided, all three must be provided.

Response Class (Status 200 OK):

Field	Description
mfaId	Unique identifier of the MFA session created for the SMS authentication request
status	Current status of the SMS authentication request and verification workflow
lockoutStatus	Indicates whether a user is currently blocked from completing MFA authentication due to security policies or exceeded verification attempts

SendVoice:Initiates a Voice authentication request

Call `api/mfa/sendVoice` to initiates a voice authentication request.

Request Body Parameters:

Parameter	Description
messageName	Name of the message template to be used for MFA modes that use message template.
cultureCode	Culture code to be used for MFA modes that use culture code. (Currently supported CC : ar- Arabic, de- German, es: Spanish, fr: French, id-ID: Indonesia, it: Italian, ja: Japanese, nl: Dutch, pt: Portuguese, ru: Russian, sv: Swedish, ta: tamil, th: thai, vi: Vietnamese, zh: chinese)
countryCode	Sms country code(s) SMS MFA mode. Multiple numbers can be separated by " " (called as vertical bar or pipe line). This should match the SMS phone numbers count.

PhoneNumber	Sms phone number(s) SMS MFA mode. Multiple numbers can be separated by " "
allowInternationalCalls	Allow international calls flag for SMS and voice MFA modes.(deprecated)
clientIPAddress	Client IP address to use to create MFA session for all MFA modes.
clientUserAgent	Client user agent to use to create MFA session for all MFA modes.
clientUserId	Client user id to use to create MFA session for all MFA modes.
messageSubstitutions	Allows recipient-specific placeholder values to be dynamically replaced in a shared email template, enabling personalized bulk email delivery using a single Message Profile and Template.
allowVoip	Determines whether MFA SMS or voice call verification can be sent to a VoIP phone number; when disabled, SMS and call delivery to VoIP numbers is not permitted.
product	List one or more WK products using this service instance. Used for itemized tracking on usage reports. Can enter upto 256 characters.
division	Division of the WK product(s) using this service instance. Allowed values: TAA, CPESG, LR, FCC, HEALTH, DXG, OTHER.
billingCode	Used to calculate monthly chargeback of SMS costs. This will generally be a WK cost center. Can enter upto 256 characters.

Note: These fields (product, division & billingCode) are collectively optional but conditionally required—if any one of product, division, or billingCode is provided, all three must be provided.

Response Class (Status 200 OK):

Field	Description
mfaId	Unique identifier of the MFA session created for the voice authentication request
status	Current status of the voice authentication request and verification workflow.
lockoutStatus	Indicates whether a user is currently blocked from completing MFA authentication due to security policies or exceeded verification attempts

Authenticator App Pairing Prompt Dialog

Create Authenticator App Pairing Prompt session

Call `client.CreateAuthPairingPrompt()` method to create WK Authenticator or Google authenticator Pairing Prompt UI.

Request Body Parameters:

Parameter	Description
-----------	-------------

<code>clientId</code>	Client user id to use to create WK or Google authenticator key.
<code>applicationName</code>	Name of the application.
<code>cultureCode</code>	Culture code to be used for App Pairing Prompt dialog UI.
<code>clientIpAddress</code>	Client IP address to use in tracing logs.
<code>clientUserAgent</code>	Client user agent to use in tracing logs.
<code>redirectUri</code>	URL to redirect to after completion of App Pairing Prompt dialog.
<code>redirectOnErrorUri</code>	URL to redirect to in case of error in App Pairing Prompt dialog.
<code>bannerAttachmentName</code>	This is the attachment name of the logo image file which is added as attachments in the MFA instance.
<code>stylesheetAttachmentName</code>	This is the attachment name of the style sheet file which is used in the MFA prompt UI.
<code>defaultPairingMethod</code>	Authenticator type selected by default.
<code>authPairMode</code>	By default value of this field is 'IDaaS'.

Response Class (Status 200 OK):

Field	Description
<code>mfaUiUrl</code>	URL for App Pairing Prompt dialog. This URL is valid for 1 minute

Within 1 minute the client browser should be redirected to the URL specified by response (`mfaUiUrl`).

Authenticator App Pairing Prompt dialog Redirect Scenarios

Successfully completes Authenticator App pairing

After the user performs app pairing, the Authenticator App Pairing Prompt dialog redirects the browser to **`redirectUri`** URL with added query string parameter:

- `completionId` : Id of completion of the Authenticator App Pairing Prompt dialog.

User cancels the Authenticator app pairing dialog

In case of user canceling the dialog, Authenticator App Pairing Prompt dialog redirects the browser to **`redirectOnErrorUri`** URL with the querystring `errorCode=Cancelled`

Verify Authenticator App Pairing Prompt session

Call `client.ValidateAppPairingCompletionToken()` method.

Query Parameters:

Parameter	Description
<code>completionId</code>	App Pairing Prompt Dialog completion Id.

<code>clientId</code> (optional)	Client user id used in <code>CreateAppPairingPrompt()</code> method.
----------------------------------	--

Response Class (Status 200 OK):

Field	Description
<code>status</code>	MFA Status of the App Pairing Prompt Dialog.

Integrating Email, SMS, and Voice to Custom UI

Email MFA

1. Call `client.SendEmail()` to start the MFA process.
2. Let the client enter verification code received in the Email.
3. Call `client.VerifyCode()` to verify the code and complete the MFA process.

[!NOTE] For the same MFA session, `VerifyCode()` method can only be called once.

SMS MFA

1. Call `client.SendSMS()` to start the MFA process.
2. Let the client enter verification code received in the SMS.
3. Call `client.VerifyCode()` to verify the code and complete the MFA process.

[!NOTE] For the same MFA session, `VerifyCode()` method can only be called once.

Voice MFA

1. Azure MFA SDK (Default)

By default we support Azure MFA SDK for performing voice MFA. In this, user needs to press # key of the phone in order to validate his identity. Following calls are needed for integrating the voice MFA:

1. Call `client.SendVoice()` to start the MFA process.
2. Once the user answers the phone call and confirms by pressing #, call `client.GetStatus()` to verify the status of the MFA process.

2. Twilio Voice MFA

In this, apart from the standard # key Voice MFA we also support OTP code being specified in the voice call which user needs to enter while performing voice MFA. The details of it can be found below:

Scenario 1:

When the "Voice Greeting Text" field of culture has `{{VerificationCode}}` then `SendVoice` returns `Status = VerificationPending`

Here we have to call `client.SendVoice()` and then `client.VerifyCode()` to verify the code and complete the MFA process.

Manage	MESSAGE DETAILS	
Settings	Name	Default
Messages	Culture Code	en-US
Attachments	SMS Text	Your sign-in verification code is: {{verificationCode}}
Users	SMS From Phone Number	
Support	Voice Greeting Text	Your sign-in verification code is: {{verificationCode}}
Generate Access Code	Voice Acceptance Text	you are successfully verified
MFA Logs	Voice From Phone Number	
MFA Email Logs	Email From	noreply@wolterskluwer.com
View	Email Subject	Verify your sign-in
Dashboard		
Audit		
Test		
Validate		

Scenario 2:

When the "Voice Greeting Text" field of culture does not have `{{VerificationCode}}` then it is verified based on # key and SendVoice returns `Status = InProgress`

Here we have to call SendVoice and then GetStatus API.

Manage	MESSAGE DETAILS
Settings	Name Default
Messages	Culture Code en-US
Attachments	SMS Text Your sign-in verification code is: {{verificationCode}}
Users	SMS From Phone Number
Support	Voice Greeting Text Welcome to Wolters Kluwer authentication System. Please press # key to authenticate.
Generate Access Code	Voice Acceptance Text you are successfully verified
MFA Logs	Voice From Phone Number
MFA Email Logs	Email From noreply@wolterskluwer.com
View	Email Subject Verify your sign-in
Dashboard	
Audit	
Test	
Validate	

Creating and verifying temporary access codes for custom scenarios

Create access code

You can develop an option to call tech support or develop utilities for the firm to provide support in case a user cannot complete MFA. After the user proved his/her identity, the tech support person can create a temporary access code that can be used by the user to perform MFA. The application may develop support utilities using the following APIs:

- `client.CreateAccessCode()` method to create access code
- `client.VerifyAccessCode()` method to verify access code

Google Authenticator (RFC 6238)

Add new Google Authenticator account

Using `userId` which identifies the user for your system (user Email or database Guid), call `client.GetTotpKey()` method to create Google Authenticator key.

`client.GenerateTotpKey()` method can also be used to generate key and QR code.

Create text to identify the Google Authenticator account to the user, it will be displayed in the Google Authenticator UI. The user may have other Google Authenticator accounts, so it is a good idea to use Organization, product, application name as part of this text: suggested format: `{Organization_Name}:{User_Name}` or `{AppName}:{User_Email}`

Call `client.CreateQrCode()` to generate QRCode image data.

Display the key and the QRCode image to the user and ask the user to use Google Authenticator to scan QR code to create new account. The user may choose to manually enter the key in Google Authenticator.

[!NOTE] Each call to `GetTotpKey()` invalidates previous key and creates a new key.

The key can be temporarily disabled and then enabled using `SetTotpEnabled()` method. This is for situations when the user lost mobile device with configured Google Authenticator account.

Verify Google Authenticator code

Call `client.VerifyTotp()` method to verify the code for the specified user id.

[!NOTE] Each code can only be verified once: calling `VerifyTotp()` again with the same code will always return "fail" status.

Optionally, you can use `StartTotpSession()` and `VerifyTotpSession()` service methods. This way you will have MFA session id for use with registered devices feature, as well as the pattern for TOTP MFA will be the same as for other MFA modes.

WK Authenticator

Add new WK Authenticator account

Using `userId` which identifies the user for your system (user Email or database Guid), call `client.GenerateTotpKey()` method to create WK Authenticator key and QR code. The key and QR code are the same as for Google Authenticator, but also have `wkmfaurl` and `wkmfauserguid` QR code parameters that will be used in WK authenticator MFA methods. Use `wkmfaurl` as server base URL for all client application HTTP calls to the server.

Display the key and the QRCode image to the user and ask the user to use WK Authenticator to scan QR code to create new account.

[!NOTE] Each call to `GetTotpKey()` or `GenerateTotpKey()` invalidates previous key and creates a new key.

The key can be temporarily disabled and then enabled using `SetTotpEnabled()` method. This is for situations when the user lost mobile device with configured WK Authenticator account.

Offline mode

In the event that client application doesn't have access to the server, Google Authenticator methods (asking the user to manually enter TOTP) can be used to verify MFA session.

Start WK Authenticator session

In the server application call `client.StartTotpWsSession()` method to start MFA session.

Get List of WK Authenticator sessions

In the client application call anonymous HTTP GET request to `api/mfa/getTotpWsSession/{userGuid}` to get the latest TOTPWS MFA session id started with `StartTotpWsSession()` method call.

Ask the user to verify MFA session Id (by showing MFA id on the server and client applications) and either confirm or deny MFA session.

Confirm or deny WK Authenticator session

In the client application call anonymous HTTP GET request to `api/mfa/confirmTotpWsSession/{userGuid}/{sessionGuid}/{totp}` with optional (default is true) parameter "accept" to confirm or deny the session.

Other Authenticators

Add new Other Authenticators account

Users can also use third party authenticators that support RFC 6238 standard. The integration steps are the same as for Google Authenticator.

Monitoring MFA service availability

To verify MFA service availability, perform the following web request:

```
1 https://{base_url}/healthmonitor/laststatus?client=YourApplicationName
```

client query string parameter can be anything to identify your application, but using your MFA service instance name is preferable.

Guidelines for adding an MFA flow using custom UI

After the user authenticates using username / password, using whatever logic is needed, determine if the user needs MFA and what MFA modes are available for the user. The logic can be based on: "remember me" token, organization policy, availability of mobile phone number and user agreement to use it for MFA, information about whether the user configured Google Authenticator, random check, etc.

MFA Service has method `CheckIfTotpKeyExists()` that will return true if `GetTotpKey()` method was previously called for this user.

The server should return to the client application the list of MFA modes, each with Id that will let server application figure out which MFA mode the user selected, as well as which parameter to use: for example, the user may have several configured phone numbers for voice MFA. Using Ids instead of actual numbers will prevent the (untrusted) client application from supplying different phone number, thus defeating MFA.

The server should not send complete phone number to the client, but provide enough info so that the user can make the selection. For example: instead of "Voice call to 555.555.1234" the server should send to client and the user should see "Voice Call to ***.***.1234". Or even better "Voice Call to home phone", where "home phone" is something that the user entered when allowing the use of his/her phone number for MFA.

For Google Authenticator, simply direct the client application to a page where the user can enter the verification code.

For Email, SMS and Voice MFA modes, start MFA session and direct the client application to the page where the user can enter verification code or simply press a button after completing MFA voice call. Keep MFA session Id on the server or keep it in cryptographically signed portion of server / client exchange. For example: if using signed authentication token with claims, add claim `"MFAID" = "{mfa_session_Id}"` . This way the server can verify that the specified MFA session was actually started for the authenticated user. This is to prevent a hacker from using MFA session ID from one user to authenticate another user.

After the user enters verification code or, in case of voice MFA, presses "Complete MFA" button, the server verifies that the user specified correct MFA session id and calls `VerifyCode()` for SMS and Email, `GetStatus()` for Voice or `VerifyTotp()` for Google Authenticator MFA mode.

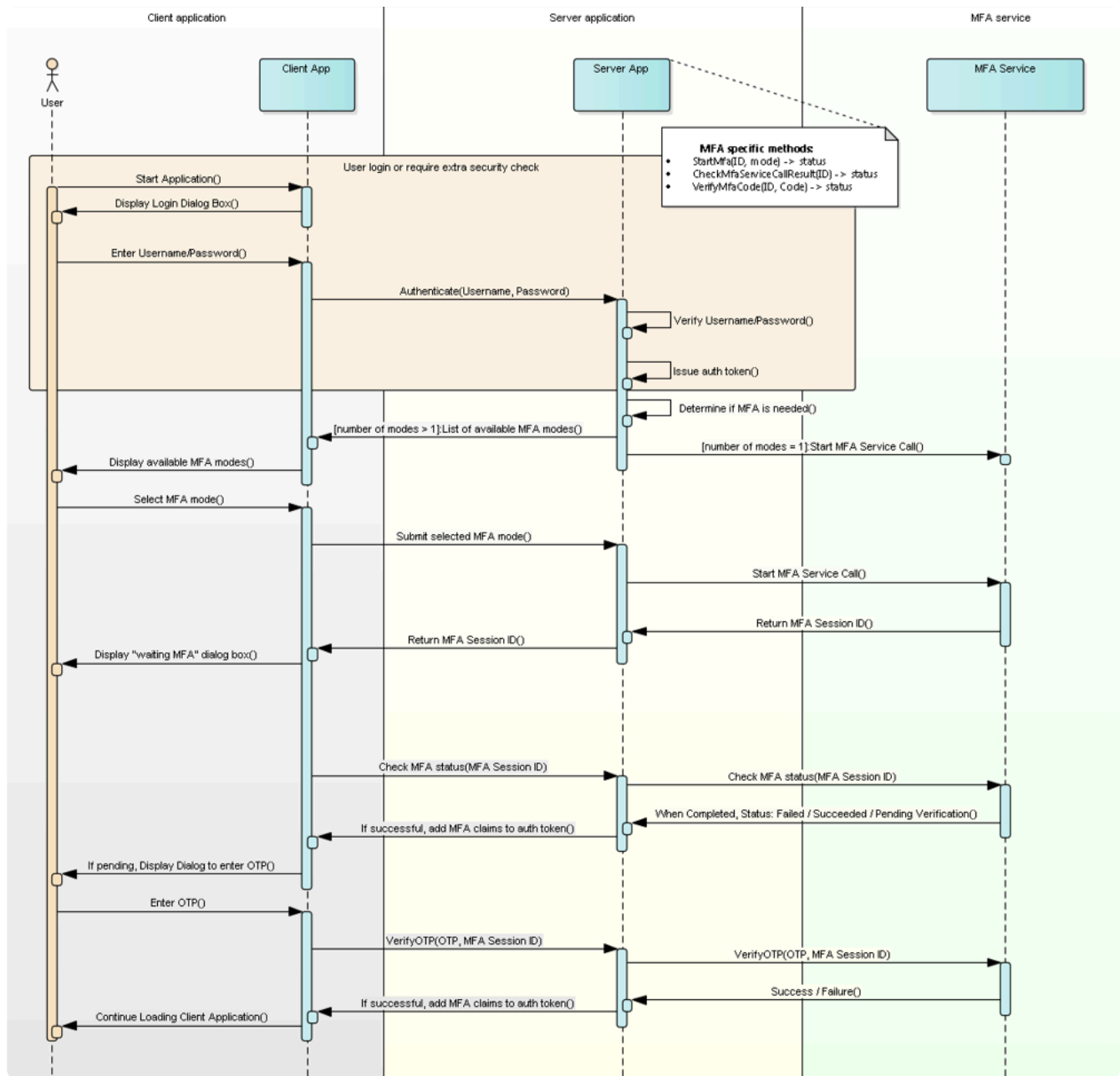
If MFA was successful, set flag / claim that MFA succeeded, date time is recommended for this purpose.

Registered Devices ("Remember me") feature

If "remember me" (registered devices) feature is needed, to minimize calls to MFA service and to simplify user experience, then set the cookie (data that the client application will store and provide with each call) to signed token with information when MFA was performed. `CreateRegisteredDevice()` and `VerifyRegisteredDevice()` service methods create and verify such token (DeviceCredential). `ReadRegisteredDevices()` and `RevokeRegisteredDevice()` service methods help manage "Registered Devices" feature.

It is recommended to use "remember me" functionality only for regular user functionality. Whenever administrator (for example: updating user phone number) functionality is needed, then that part of application may require MFA performed in the current session or even, in case of Google Authenticator setup, require MFA performed in the current session within the last 5 minutes. If using claims identity, set "MFA Performed" and "RememberMe Verified" claims separately.

Here's suggested client / server / MFA service interaction diagram:



SignalR Notifications

For Voice and WK Authenticator MFA modes, the MFA service invokes SignalR based client events to notify the status of the MFA process. This allows integrating applications to automatically detect when an out-of-band authentication is completed. The Client applications can connect to the Hub and register to the events.

The Client applications using JavaScript to connect SignalR Hub can refer to: [SignalR JavaScript Client Guide](#)

The Client applications using .NET to connect SignalR Hub can refer to: [SignalR .NET Client Guide](#)

The MFA Service SignalR Hub is available at the following URL: `https://<MFAServiceURL>/signalr/hubs`

The MFA Notifications are available in the Hub name **"mfaHub"**

The SignalR Hub to send notification to the appropriate MFA session, the client application needs to provide the MFA Session ID to the Hub by calling the server method "subscribe" and pass the MFA session id. After the completion

of MFA session, the Client applications can unsubscribe the notification for the MFA session Id as it's no longer needed by calling the server method "unsubscribe" and pass the MFA session id.

In event of the status change for the Voice and WK Authentication MFA modes, the Hub invokes the client event "loginCompleted" with the Boolean value as parameter. The "True" value denotes the MFA session is validated and approved. The "False" value denotes the MFA session is validated and denied.

Important: After the SignalR notification received from the SignalR Hub, the consuming applications MUST validate the MFA Session Status with the MFA Service to complete MFA validation.

Sample JavaScript code which refers the dynamically generated hub proxy:

```
1  var hubProxy = $.connection.mfaHub;
2
3  $.connection.hub.url = "https://<MFAServiceURL>/signalr/hubs";
4
5  // Add MfaId to SignalR requests.
6  $.connection.hub.qs = {
7      mfaSessionId: function () {
8          var mfaSessionValue = <MFA Session Id>;
9          return (typeof mfaSessionValue === "undefined" || mfaSessionValue === null)
10             ? ""
11             : mfaSessionValue;
12      }
13  };
14
15  $.connection.hub.start().done(function () {
16      hubProxy.server.subscribe(<MFA Session Id>);
17  });
18
19  hubProxy.client.loginCompleted = function (status) {
20      if (status) {
21          // Do Something - MFA approved
22      } else {
23          // Do Something - MFA denied
24      }
25
26      hubProxy.server.unsubscribe(<MFA Session Id>);
27      $.connection.hub.stop();
28  };
29
30  // The MFA Session is valid only for 5 Mins. In case of no response from the User,
31  // The Client should end the MFA Session and begin new session.
32  setTimeout(function () {
33      timedOutAutoLogin();
34  }, 300000);
35
36  function timedOutAutoLogin() {
37      hubProxy.server.unsubscribe(<MFA Session Id>);
38      $.connection.hub.stop();
39  }
```

To consume the dynamically generated proxy from the MFA service URL, the client application should allow the SignalR hubs URL in the **CORS** policy of the application.

Wolters Kluwer Authenticator - Web Authentication

Auth Pairing Prompt

MFA Prompt UI also includes Wolters Kluwer Authenticator support. Applications who are using MFA Prompt UI now can register the users for Wolters Kluwer Authenticator using AuthPairing Prompt.

Applications can create Auth Pairing Prompt using the following MFA WebAPI Method.

POST

/api/mfa/createAuthPairingPrompt

Creates the authentication pairing prompt.

Parameters

Try it out

No parameters

Request body

application/json-patch+json

The request.

Example Value | Schema

```

{
  "clientId": "username@wolterskluwer.com",
  "applicationName": "TAA MFA Application",
  "cultureCode": "en-US",
  "clientIdAddress": "123.123.234.234",
  "clientIdAgent": "Mozilla/5.0 (Windows NT 10.0; WOW64; Trident/7.0; rv:11.0) like Gecko",
  "bannerAttachmentName": "bannerAttachment",
  "stylesheetAttachmentName": "stylesheetAttachment",
  "redirectUri": "https://httpbin.org/get",
  "redirectUriMethod": "https://httpbin.org/get",
  "defaultPairingMethod": "WKA",
  "authPairMode": "IDAAS"
}

```

Responses

Code	Description	Links
200	OK	No links

Media type

text/plain

Controls Accept header

Example Value | Schema

```

{
  "authPairingUri": "https://suswebapi.chaccess.com/public/apppairingprompt/en-US?appPairPromptId=f7d790d274da4f4844c6d7d7cac16297"
}

```

The Auth Pairing Prompt UI displays the appropriate UI based on the `defaultPairingMethod` value. The valid value is:

- WKA - for Wolters Kluwer Authenticator only

The Auth Pairing Prompt returns a Unique URI to the applications to redirect the users to register for the Wolters Kluwer Authenticator.

Wolters Kluwer Authenticator will prompt the users to follow the directions displayed on the screen.

Users can select the appropriate device and click Continue to get the App Information and QR Code.

Wolters Kluwer

2-Step Verification

Wolters Kluwer Authenticator

Before you can use the application for verification, we need to do a one time setup

What type of phone do you have?

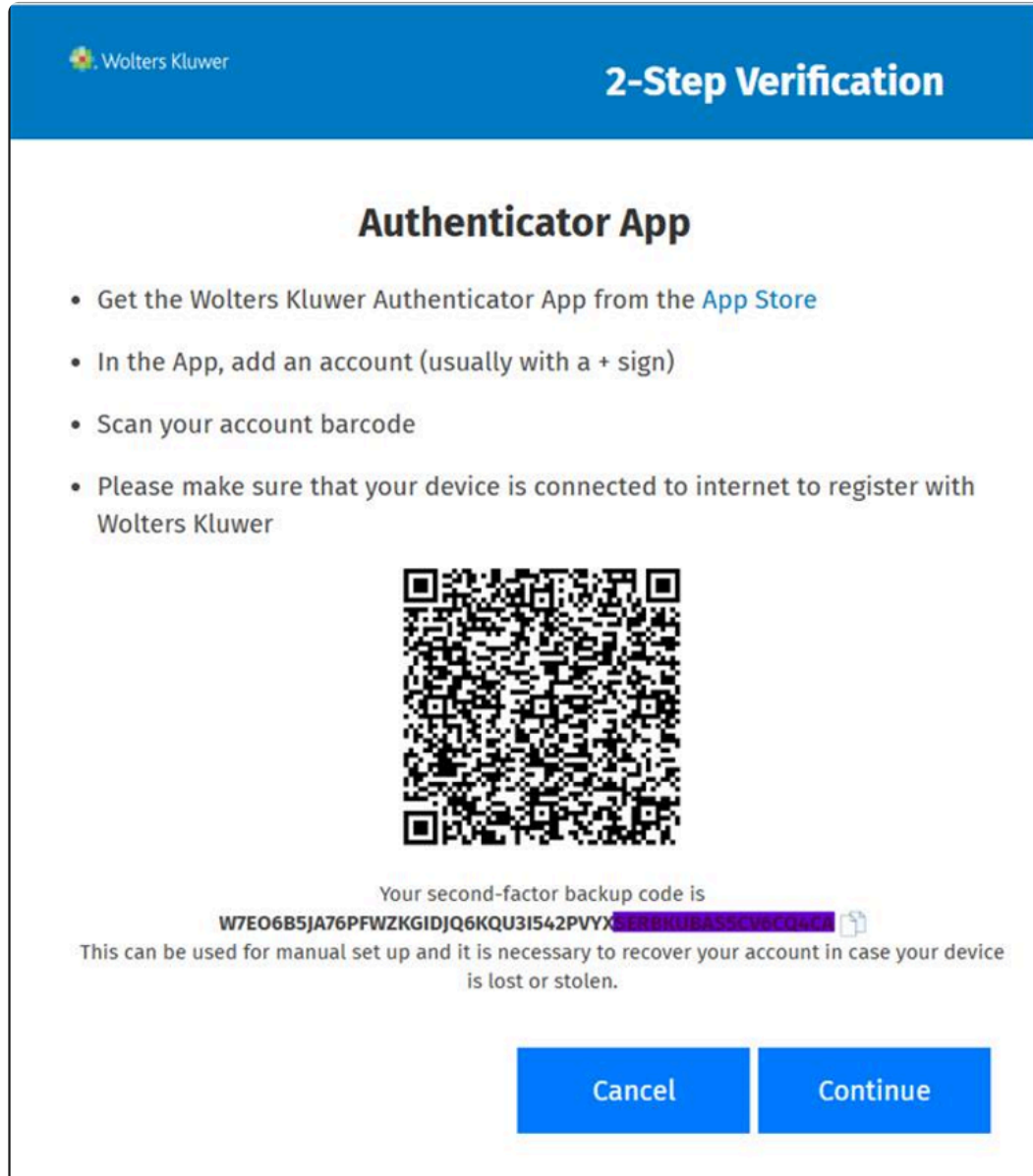
iPhone

Android

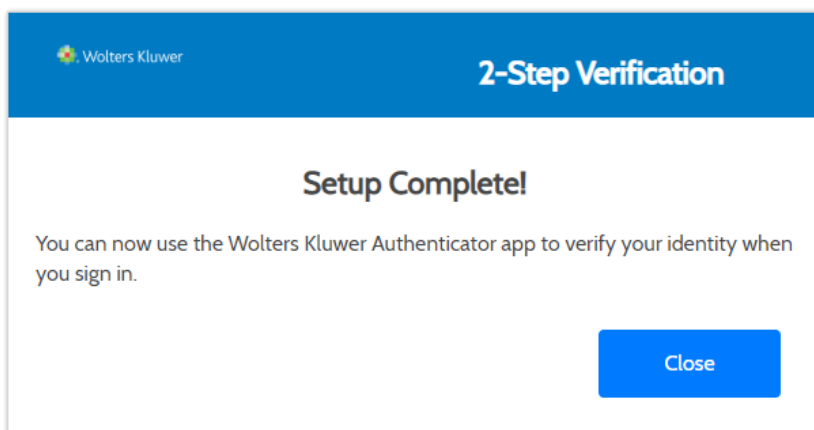
Cancel

Continue

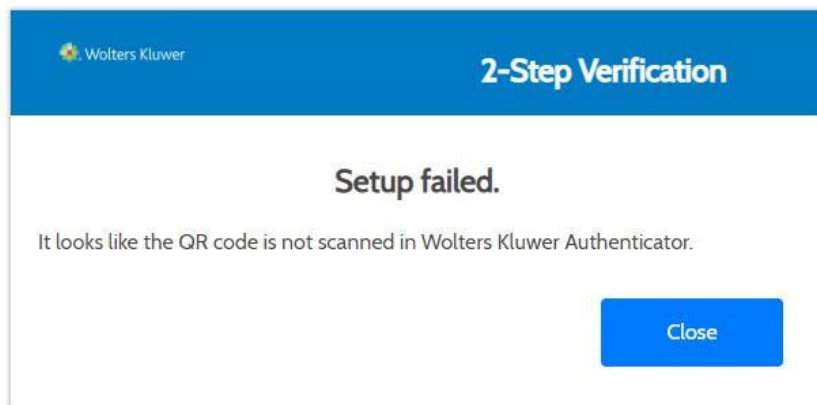
Selecting iPhone will render UI with App store URL to download the application and QR Code to add the account in Authenticator app.



After Installing the App and Scanning the QR Code in the App, click Continue. Successful completion of the Pairing will render the following page and clicking close will redirect to the given redirectUrl with authPairingCompletionId.



Failure to install and scan the QR Code will result the Pairing to fail and results in the following UI:



Clicking on the Close button redirects back the application (redirectUrl mentioned in the AuthPairingPrompt) with `errorCode=NOTPAIRED`

Example: <https://httpbin.org/get?errorCode=NOTPAIRED>

Verify Auth Pairing Completion

Applications can call the MFA Web API method `validateAuthPairingCompletion` to verify the `authPairingCompletionId`.

GET /api/mfa/validateAuthPairingCompletion Validates the authentication pairing completion token.

Parameters

Name	Description
completionId string (query)	The completion identifier. completionId
clientId string (query)	The client user identifier. clientId

Responses

Code	Description	Links
200	OK Media type: text/plain Controls Accept header: ☐ Example Value Schema <pre>{ "status": "Success" }</pre>	No links

MFA Verify Dialog

The MFA Verify Dialog provides a standard user interface that applications can use to allow users to confirm their MFA options.

To integrate this functionality, the application passes in the MFA parameters with one of the MFA mode supported as well as a redirectUrl for continuation and gets back a **mfaUiUrl** with an embedded session ID. Then the application directs the user to **mfaUiUrl**. The MFAVerify initiates the new MFA Session for the given MFA Mode and prompts the user to enter the code which is sent to the user for the given MFA Mode. The application makes a

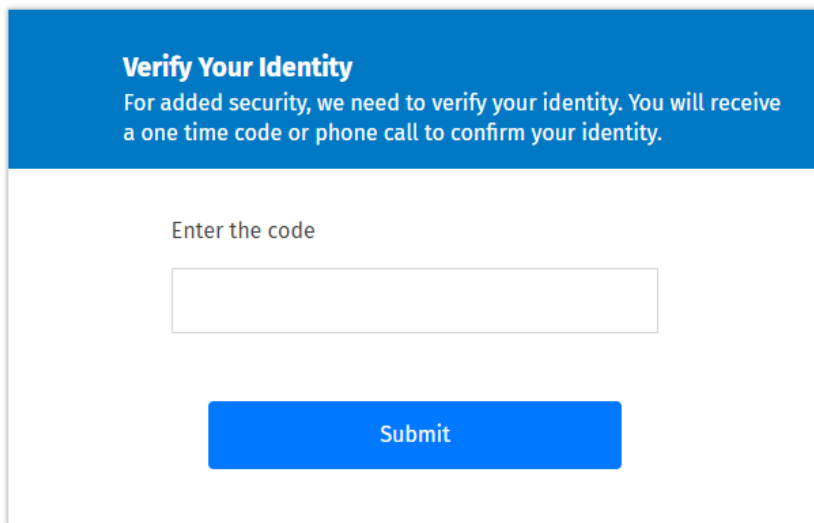
separate call to `validateMfaCompletion` and can ensure that the **mfaCompletionId** was valid and gets back some information it can choose to record in audit logs on how the user completed the challenge.

In the current release, the following features of MFA are supported with **createMfaVerify**:

1. Email
2. SMS
3. Voice Call, with SignalR detection of the out-of-band authentication being completed

Sample UI:

Unlike the MFAPrompt, the MFA session will be initiated for the given MFA mode and user will be prompted to confirm the MFA Mode.

A sample user interface for MFA verification. It features a blue header with the text "Verify Your Identity" and a sub-header "For added security, we need to verify your identity. You will receive a one time code or phone call to confirm your identity." Below the header is a white box with the label "Enter the code" and a text input field. At the bottom of the white box is a blue "Submit" button.

Create MFA Verify session

Call `client.CreateMfaVerify()` method to create MFA Verify session.

Request Body Parameters:

Parameter	Description
<code>messageName</code>	Name of the message template to be used for MFA modes that use message template.
<code>cultureCode</code>	Culture code to be used for MFA modes that use culture code. (Currently supported CC : ar- Arabic, de- German, es: Spanish, fr: French, id-ID: Indonesia, it: Italian, ja: Japanese, nl: Dutch, pt: Portuguese, ru: Russian, sv: Swedish, ta: tamil, th: thai, vi: Vietnamese, zh: chinese)
<code>emailAddress</code>	Email address to be used for email MFA mode. Multiple E-mail id(s) can be separated by " "
<code>smsCountryCode</code>	Sms country code(s) SMS MFA mode. Multiple numbers can be separated by " " (called as vertical bar or pipe line). This should match the SMS phone numbers count.
<code>smsPhoneNumber</code>	Sms phone number(s) SMS MFA mode. Multiple numbers can be separated by " "

allowInternationalCalls	Allow international calls flag for SMS and voice MFA modes. (Deprecated)
voiceCountryCode	Voice country code Voice MFA mode. Multiple numbers can be separated by " "
voicePhoneNumber	Voice phone number Voice MFA mode. Multiple numbers can be separated by " "
attachments	An optional array of attachment names used in Email MFA mode.
messageSubstitutions	Message substitutions Json object used in Email and SMS MFA modes.
clientIPAddress	Client IP address to use to create MFA session for all MFA modes.
clientUserAgent	Client user agent to use to create MFA session for all MFA modes.
clientId	Client user id to use to create MFA session for all MFA modes.
redirectUri	URL to redirect to after completion of MFA.
redirectOnErrorUri	URL to redirect to in case of error in MFA Prompt.
bannerAttachmentName	This is the attachment name of the logo image file which is added as attachments in the MFA instance.
stylesheetAttachmentName	This is the attachment name of the style sheet file which is used in the MFA prompt UI.
mfaMode	'email', 'sms', or 'voice'

Response Class (Status 200 OK):

Field	Description
mfaUiUrl	URL for MFA Verify dialog. This URL is valid for 1 minute

Within 1 minute the client browser should be redirected to the URL specified by response (mfaUiUrl).

The user will have 5 minutes (**Session Timeout** value in Service instance settings) to confirm the MFA option.

MFA Verify UI Custom Stylesheet

It works same way as mentioned in the section [MFA Prompt UI Custom Stylesheet](#)

MFA Verify UI Custom Banner Logo Image

It works same way as mentioned in the section [MFA Prompt UI Custom Banner Logo Image](#)

MFA Verify UI Redirect Scenarios

It works same way as mentioned in the section [MFA Prompt UI Redirect Scenarios](#)

Verify MFA Verify session

It works same way as mentioned in the section [Verify MFA Prompt session](#)

MFA Short Circuit

With MFA Short Circuit configuration switch, we can bypass MFA for products by modifying their configuration in the event of a network disruption or period of MFA problems. Products that are using MFA Client SDK can use this feature.

To enable Short Circuit, add configuration key `mfa:ShortCircuitService` and set it to `true` in your application. Set it to `false` to disable it.

The MFA Prompt UI will be bypassed and below are APIs' responses when MFA Short Circuit is configured:

SendEmail, SendSMS, SendVoice:

```
1 SendMfaResponse
2 {
3     MfaId = "00000000-0000-0000-0000-000000000000",
4     Status = InProgress,
5     LockoutStatus = None
6 }
```

VerifyCode:

```
1 WebApiMfaStatus = Success
```

CreateMfaPrompt:

```
1 CreateMfaPromptResponse
2 {
3     MfaUiUrl = redirectUri + "&mfaCompletionId=00000000-0000-0000-0000-000000000000"
4 }
```

ValidateMfaCompletionToken:

```
1 ValidateMfaCompletionTokenResponse
2 {
3     Status = Success,
4     MfaMode = "ShortCircuit"
5 }
```

Track SMS/Voice status

MFA has an Admin API `/api/mfa/admin/serviceInstances/{name}/audit`

GET /api/mfa/admin/serviceInstances/{name}/audit Returns session log details for the specified service instance.

Returns session log details for the specified service instance. Requires Service Owner authentication.

Parameters

Name
Description

name
Service instance Name

string
(path)

startTimeUtc
if specified, then return records created after.

string
(query)

endTimeUtc
if specified, then return records created before.

string
(query)

status
if specified, then filter by comma separated list of statuses.

string
(query)

authenticationType
if specified, then filter by comma separated list of authentication types.

string
(query)

messageName
if specified, then filter by name of messages.

string
(query)

cultureCode
if specified, then filter by name of message culture.

string
(query)

clientIpAddress
if specified, then filter by client IP address.

string
(query)

clientId
if specified, then filter by client user id.

string
(query)

clientUserEmail
if specified, then filter by client user email.

string
(query)

limitRecordCount
if specified, then the maximum of records to return. This number can be between 0 and 1000

integer(\$int32)
(query)

Execute

Responses

Code
Description
Links

200
OK
No links

Media type
text/plain

Controls Accept header.

Example Value | Schema

```

{
  "authenticationType": "Email",
  "serviceInstanceName": "ServiceInstanceName",
  "status": "Processed",
  "messageName": "Default",
  "cultureCode": "en-US",
  "retryCount": 5,
  "clientIpAddress": "123.123.234.234",
  "clientId": "username@alterskinner.com",
  "clientUserEmail": "username@alterskinner.com",
  "authenticationParameters": "<html version='1.0' encoding='utf-16'><Email Address='username@alterskinner.com' />",
  "creationTimestamp": "2020-04-23T08:31:24.4964252Z",
  "expirationTimestamp": "2020-04-23T08:31:24.4964252Z",
  "clientUserAgent": "Mozilla/5.0 (Windows NT 10.0; Win64; Trident/7.0; rv:11.0) like Gecko",
  "deliveryInfo": [ { "event": "delivered", "email": "username@alterskinner.com", } ],
  "messageSubstitutions": { "user": { "code": "123456", "state": "GA", "address": "123 Street" } }
}

```

You need to pass filter parameters like instance name, start time, end time, ClientUserId, etc. This API will return a list of logs and each log has a property "deliveryInfo" which stores all information related to SMS/Voice being sent like Status (Queued, Sent, Delivered...), SMS from, SMS to, etc. in JSON format.

Below is the sample deliveryInfo data format:

For Plivo:

```

1 [
2   {
3     "ServiceType": "MFA",

```

```

4    "PlivoSmsFrom": "1000000000",
5    "PlivoMessageUUID": "e6939838-5e37-11eb-bfe8-0242ac110004",
6    "PlivoSmsStatus": "delivered",
7    "PlivoSmsTo": "918087601872",
8    "ErrorCode": null,
9    "ParentMessageUUID": "e6939838-5e37-11eb-bfe8-0242ac110004",
10   "PowerpackUUID": null,
11   "MessageTime": "2021-01-24 11:32:54.957701",
12   "QueuedTime": "2021-01-24 11:32:54.973325",
13   "SentTime": "2021-01-24 11:32:55.149025",
14   "TotalAmount": "0.021",
15   "TotalRate": "0.02100",
16   "Units": "1"
17 }
18 ]

```

For Twilio:

```

1 [
2   {
3     "ServiceType": "MFA",
4     "TwilioAccountSid": "AC3a2ea6e66c143e6e3a1ac3f463c9567f",
5     "TwilioApiVersion": "2010-04-01",
6     "TwilioFrom": "+13122573476",
7     "DeliveryMessageSid": "SMfa1585a94cf9429788f32d3186244d9d",
8     "TwilioMessageStatus": "delivered",
9     "TwilioSmsSid": "SMfa1585a94cf9429788f32d3186244d9d",
10    "TwilioSmsStatus": "delivered",
11    "TwilioTo": "+918087601872"
12  }
13 ]

```

You can also access these logs using SWS Portal from MFA service instance:

SERVICE INSTANCE: MFA_130520_01

Manage

- Settings
- Messages
- Attachments
- Users

Support

- Generate Access Code
- MFA Logs
- MFA Email Logs

View

- Dashboard
- Audit

Test

- Validate

From (UTC): 02/01/2021 6:45 PM To (UTC): 02/03/2021 6:45 PM

User Id: rohit.rusla@wolterskluwer.com

Search

MFA Initiated Timestamp	Authentication Type	Status	Client IP Address	Delivery Info
2/2/2021 2:19:04 PM	Email	VerificationPending	123.123.234.234	
2/2/2021 2:19:03 PM	Email	InProgress	123.123.234.234	
2/2/2021 12:16:44 PM	SMS	VerificationPending	123.123.234.234	[{"ServiceType": "MFA", "TwilioAccountSid": "AC3a2ea6e66c143e6e3a1ac3f463c9567f", "TwilioApiVersion": "2010-04-01", "TwilioFrom": "+13122573476", "DeliveryMessageSid": "SMfa1585a94cf9429788f32d3186244d9d", "TwilioMessageStatus": "delivered", "TwilioSmsSid": "SMfa1585a94cf9429788f32d3186244d9d", "TwilioSmsStatus": "delivered", "TwilioTo": "+918087601872"}]
2/2/2021 12:16:41 PM	SMS	VerificationPending	123.123.234.234	[{"ServiceType": "MFA", "TwilioAccountSid": "AC3a2ea6e66c143e6e3a1ac3f463c9567f", "TwilioApiVersion": "2010-04-01", "TwilioFrom": "+13122573476", "DeliveryMessageSid": "SMfa1585a94cf9429788f32d3186244d9d", "TwilioMessageStatus": "sent", "TwilioSmsSid": "SMfa1585a94cf9429788f32d3186244d9d"}]

Upgrading from an older SDK version

Upgrade from SDK version 1.21

- NuGet package name changed to `WK.TAA.SharedWebServices.ClientSdk.NSwag.Mfa` from `WK.TAA.SharedWebServices.ClientSdk.Mfa`. For usage, see section [Installing MFA SDK](#)
- The type `SharedWebServicesClient` which was available in version 1.21 is no longer available from version 1.23. To use SDK, create a client object of `MultiFactorAuthenticationClient` in place of `SharedWebServicesClient`. For usage see section [Create client object](#)
- Because `MultiFactorAuthenticationClient()` can now be directly instantiated, methods like `CreateMFAPrompt()`, `CreateAppPairingPrompt()` and `SendEmail()` etc. can be directly called using

this object. For example, in application code something like

```
client.MultiFactorAuthentication.CreateMfaPrompt()
```

 will get replaced by

```
client.CreateMfaPrompt()
```

 . For usage see [Create MFA Prompt session](#)

Document History

Revision Date	Revised by	Summary of Changes
26th March 2026	Sameer Mangaonkar	Release 31 features
29th January 2025	SWS	Release 29 features